CS758-Lecture02

Topics

- Basic definitions and representation
- Gram-Schmidt algorithm
- Characterising lattice bases

Sources

- [1]: BOOK-NguVal10 STUB, Chapter 02 (PDF)
- [2]: COURSE-Micciancio07, Lecture 02 (PDF)

Plan

- At the end of the lecture, you will have understood:
  - The difference between vector (sub)spaces and lattices
  - Basic properties of lattice bases
- Basic definitions and representation
  - We work over an n-dimensional vector space (to be specific, the Euclidean space) equipped with the standard notion of inner product (and thus norm, i.e., $\|\bar{x}\| := \langle \bar{x}, \bar{x} \rangle = \sum_{i=1}^{n} x_i^2$)
  - "Topological" definition via discreteness
    - Topologically, a lattice is a set of "regular" "well-spaced" points in $\mathbb{R}^n$
    - **Definition 1** (see [1], Definition 1): A subset $\mathcal{S}$ of $\mathbb{R}^n$ is called discrete if for every $\bar{x} \in \mathcal{S}$, there exists a $\rho > 0$ such that

      $$B(\bar{x}, \rho) \cap \mathcal{S} = \{\bar{x}\},$$

      where $B(\bar{x}, \rho)$ is the open ball of radius ρ centred at $\bar{x}$
    - **Definition 2** (see [1], Definition 4): A lattice Λ of $\mathbb{R}^n$ is a *discrete subgroup* of $(\mathbb{R}^n, +)$. Recall that a subgroup of $(\mathbb{R}^n, +)$ is a subset of $\mathbb{R}^n$ that is also a group.
    - **PQ1**. Lattice or not?
      1. $\{0^n\}$: Yes. It is the trivial "zero" lattice
      2. $\mathbb{N}^n$: No. Although it is discrete (e.g., for $\rho = 0.5$), it is not a subgroup of $\mathbb{R}^n$
      3. $\mathbb{Z}^n$: Yes. It is the integer lattice
      4. $\mathbb{Q}^n$: No. Although it is a subgroup of $\mathbb{R}^n$, it is not discrete (HW)
      5. $\{(x_1, \cdots, x_n) \in \mathbb{Z}^n : \sum_{i=1}^{n} a_i x_i = 0 \bmod q\}$, where $a_1, \cdots, a_n, q \in \mathbb{N}$. Yes. It is discrete since any subset of a discrete set is discrete. It can also be shown to be a subgroup (HW).
    - **HW1**:

1. What happens if we flip the order of quantifiers in Definition 1? That is, require that there exists a $\rho > 0$ for every $\bar{x} \in \mathcal{S}$, such that

$$B(\bar{x}, \rho) \cap \mathcal{S} = \{\bar{x}\}?$$

2. In Definition 2, what if you require Λ to be a discrete *subspace*?
3. Can the cardinality of Λ be *uncountable*?

- "Algebraic/geometric" definition via basis
  - Algebraically/geometrically, a lattice is the set of all vectors in the "integer span" of a basic $\bar{B}$
  - **Definition 3**. Let $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_m)$ be *linearly independent* vectors in $\mathbb{R}^n$. The lattice *generated by* $\bar{B}$ is defined as

$$\Lambda(\bar{B}) := \{\bar{B}\bar{x} : \bar{x} \in \mathbb{Z}^m\} = \left\{ \sum_{i=1}^{m} x_i \bar{b}_i : x_1, \cdots, x_m \in \mathbb{Z} \right\}$$

    - **Example 1.** Consider the lattice generated by $\bar{B} := ((3,0)^t, (1,2)^t)$
    - *We* will refer to n as the *dimension* of Λ and m as its *rank*. The lattice is full rank if $m = n$.
      - Note: this terminology is not consistent across papers (e.g., Regev's lecture notes follows this)
  - **HW2.** (Every lattice according to Definition 1 has at least one basis): Let Λ be a lattice as per Definition 1. For some $m \leq n$, show that there exists m linearly independent vectors $\bar{b}_1, \cdots, \bar{b}_m \in \mathbb{R}^n$ such that Λ is generated by $\bar{b}_1, \cdots, \bar{b}_m$.

- Lattice *vs* vector space ([2])
  1. A basis for a lattice is also a basis for the vector space, but converse is not true:
     - **Example 2.** Given a basis $\bar{B}$ (e.g., $(1,0)^t$ and $(0,1)^t$) for a lattice Λ (e.g., $\mathbb{Z}^2$), $2\bar{B}$ is still a basis for the vector space, but not Λ.
  2. Vector spaces *always* admit an *orthogonal* basis and these can be efficiently computed using the Gram-Schmidt algorithm:
     - **Algorithm 1**:
       - *Input*: Basis $\bar{b}_1, \cdots, \bar{b}_m \in \mathbb{R}^n$ for some vector space $\mathcal{V}$
       - *Output*: Orthogonal basis $\bar{b}_1^*, \cdots, \bar{b}_m^* \in \mathbb{R}^n$ for $\mathcal{V}$
       - Algorithm:
         1. Set $\bar{b}_1^* := \bar{b}_1$
         2. Compute $\bar{b}_2^* := \bar{b}_2 - \pi_{\bar{b}_1^*}(\bar{b}_2)$, where $\pi_{\bar{u}}(\bar{v})$ denotes the orthogonal projection of $\bar{b}_2$ on to $\bar{b}_1^*$, i.e.,

$$\frac{\langle \bar{b}_2, \bar{b}_1^* \rangle}{\langle \bar{b}_1^*, \bar{b}_1^* \rangle} \bar{b}_1^*$$

         3. For each $i \in [3, m]$: compute $\bar{b}_i^* := \bar{b}_i - \sum_{j<i} \pi_{\bar{b}_j^*}(\bar{b}_i)$ (i.e., pick $\bar{b}_i^*$ to be orthogonal to span of all previous vectors)

4. Output $\bar{b}_1^*, \cdots, \bar{b}_m^*$

- **Example 3.** On running Gram-Schmidt algorithm on $\bar{B} := ((3,0)^t, (1,2)^t)$ results in $((3,0)^t, (0,2)^t)$
- **Fact 1 (Theorem 9 in [2]).** Gram-Schmidt algorithm runs in polynomial time.
- **PQ2.** Does Gram-Schmidt algorithm yield an orthogonal *lattice* basis?
  - Not necessarily, in Example 2: $(0,2)^t$ output above is not even in the lattice $\Lambda(\bar{B})$!
  - Where/why does it fail? The value of $\langle \bar{b}_2, \bar{b}_1^* \rangle / \langle \bar{b}_1^*, \bar{b}_1^* \rangle$ in the above equation need not be an integer.

- Characterising lattice bases
  - In general, lattice bases are *much more complex* than vector space bases! For example, some lattices do not even have an orthogonal basis!
    - **Example 4.** The lattice generated by $(2,0)^t$ and $(1,2)^t$ contains orthogonal vectors (e.g, $(0,4)$ and $(2,0)$), but it doesn't have an orthogonal basis. (HW: argue formally)
  - Looking ahead, this is exactly why lattices are interesting from an algorithmic/cryptographic point of view: several problems that are computationally easy over vector spaces (e.g., finding orthogonal basis) become computationally hard over lattices! To understand lattice bases better, let's start with the following question.
  - **PQ3.** Is the basis for vector space unique? What about lattices?
    - Basis for vector spaces need not be unique. Similarly, same lattice can have more than one basis.
    - **Example 5.**: $\bar{B}_1 := ((2,0)^T, (2,2)^T)$ and $\bar{B}_2 := ((2,0)^T, (0,-2)^T)$ are also basis of the scaled integer lattice $2\mathbb{Z}^n$.
  - Let's try to characterise all the basis of a lattice. This this end we will need two definitions: fundamental parallelepiped and determinant.
    - **Definition 4.** (Fundamental parallelepiped) Given a basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_m) \in \mathbb{R}^{n \times m}$, the *fundamental parallelepiped* associated with $\bar{B}$ is the region

$$P(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i \bar{b}_i : 0 \leq x_i < 1 \right\}$$

      - Geometrically, when $m = n$, the whole space $\mathbb{R}^n$ can be obtained by "tiling" using $P(\bar{B})$.
    - **Definition 5.** (Determinant) Let $\bar{B} \in \mathbb{R}^{n \times m}$ be a basis. The determinant of $\Lambda(\bar{B})$ is the m-dimensional volume of $P(\bar{B})$:

$$det(\Lambda(\bar{B})) = vol(P(\bar{B}))$$

      - Geometrically, it captures density of lattice points in its ambient space: if the determinant is small then the lattice is dense and vice versa

- **HW4 (Hadamard inequality).** Show that $det(\Lambda(\bar{B})) \leq \Pi_{i=1}^{m} ||\bar{b}_i||$
- **HW5 (Determinant is efficiently computable).** Let $\bar{B}^* := (\bar{b}_1^*, \cdots, \bar{b}_m^*) \in \mathbb{R}^{n \times m}$ be the basis obtained by running the Gram-Schmidt algorithm on $\bar{B}$ in Definition 5. Then

$$det(\Lambda(\bar{B})) = \Pi_{i=1}^{m} ||\bar{b}_i^*||.$$

- **HW6 (Determinant is multiplicative.)** For any $\bar{M}_1, \bar{M}_2 \in \mathbb{Z}^{m \times m}$, show that

$$det(\bar{M}_1 \bar{M}_2) = det(\bar{M}_1) det(\bar{M}_2).$$

- **Theorem 1.** ([2], Theorem 2) Let $\bar{B}_1, \bar{B}_2 \in \mathbb{R}^{n \times m}$ be two bases. Then $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$ iff there exists a unimodular matrix $\bar{U} \in \mathbb{Z}^{m \times m}$ such that $\bar{B}_2 = \bar{B}_1 \bar{U}$, where an (integer) matrix $\bar{U}$ is unimodular if $det(\bar{U}) = \pm 1$.
  - Intuitively, a unimodular matrix preserves "volume" of the fundamental parallelepiped.
  - **PQ4.** Given Theorem 1, how do you enumerate all bases for a lattice Λ?
  - **Proof of Theorem 1.**
    - We first prove the following lemma about unimodular matrices:
      - **Lemma 1.** If $\bar{U} \in \mathbb{Z}^{m \times m}$ is a unimodular matrix, then so is $\bar{U}^{-1}$
      - **Proof.** We split the proof into two parts:
        1. In the first part, we argue that $\bar{U}^{-1}$ is an *integer* matrix. To see why, parse $\bar{U}^{-1}$ as $(\bar{v}_1, \cdots, \bar{v}_m)$ and each $\bar{v}_i$ as $(v_{i,1}, \cdots, v_{i,m})^t$ and consider the equation

$$\bar{U}\bar{U}^{-1} = \bar{I}_m.$$

           Using Cramer's algorithm, the system of linear equations

$$\bar{U}\bar{v}_i = \bar{e}_i,$$

           where $\bar{e}_i$ is the ith unit vector, has a unique solution $\bar{v}_i$ where

$$v_{i,j} := \frac{det((\bar{u}_1, \cdots, \bar{u}_{i-1}, \bar{e}_i, \bar{u}_{i+1}, \cdots, \bar{v}_n))}{det(\bar{U})}.$$

           Since all the vectors in the numerator are integer, and $det(\bar{U}) = \pm 1$, we have that $\bar{v}_i$ is also an integer vector and thus $\bar{U}^{-1}$ is also an integer vector.
        2. In the second part, we argue that $det(\bar{U}^{-1}) = \pm 1$. However, this follows by the fact that determinant is multiplicative (HW6):

$$det(\bar{U}) \cdot det(\bar{U}^{-1}) = det(\bar{I}_m) = 1$$

           where $\bar{I}_m$ denotes the m-dimensional identity matrix.
    - Given Lemma 1, let's prove Theorem 1:

- $\Leftarrow$ Suppose $\bar{B}_2 = \bar{B}_1\bar{U}$ for some unimodular matrix $\bar{U}$, which implies $\Lambda(\bar{B}_2) \subseteq \Lambda(\bar{B}_1)$. Since $\bar{B}_1 = \bar{B}_2\bar{U}^{-1}$ and since $\bar{U}^{-1}$ is an integer matrix by Lemma 1, we also have $\Lambda(\bar{B}_1) \subseteq \Lambda(\bar{B}_2)$. Together, $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$.
- $\Rightarrow$ Suppose $\bar{B}_1 := (\bar{b}_{1,1}, \cdots, \bar{b}_{1,m}), \bar{B}_2 := (\bar{b}_{2,1}, \cdots, \bar{b}_{2,m}) \in \mathbb{R}^{n\times m}$ be two bases such that $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$. This implies, in particular, that $\bar{b}_{1,1}, \cdots, \bar{b}_{1,m} \in \Lambda(\bar{B}_2)$ and vice versa. Thus, by definition of lattices, there exists two *integer* matrices $\bar{M}_1, \bar{M}_2 \in \mathbb{Z}^{m\times m}$ such that $\bar{B}_2 = \bar{B}_1\bar{M}_1$ and $\bar{B}_1 = \bar{B}_2\bar{M}_2$. Combining the two equations, we have

  $$\bar{B}_1 = \bar{B}_1\bar{M}_1\bar{M}_2$$

  and thus $\bar{B}_1(\bar{I}_m - \bar{M}_1\bar{M}_2) = \bar{0}_{m\times m}$, where $\bar{0}_{m\times m}$ is the $m \times m$ matrix with zero entries. Since $\bar{B}_1$ is a basis and its column vectors are linearly independent, it follows that $\bar{M}_1\bar{M}_2 = \bar{I}_m$ and thus

  $$det(\bar{M}_1\bar{M}_2) = 1.$$

  Since $\bar{M}_1$ and $\bar{M}_2$ are both integer matrices, their determinants are also integers and thus only way the above equation is satisfied (since $det(\bar{M}_1\bar{M}_2) = det(\bar{M}_1)det(\bar{M}_2)$) is when $det(\bar{M}_1) = det(\bar{M}_2) = \pm 1$.

- **HW7.** Show that if $\bar{B}_1$ and $\bar{B}_2$ are bases of the same lattice, then $det(\Lambda(\bar{B}_1)) = det(\Lambda(\bar{B}_2)$.
  - Hint: use Theorem 1.

2026s-CS758-Lecture03

Topics

- Characterising lattice bases using unimodular matrices
- Lattice determinant

  Sources
- [1]: COURSE-Micciancio07, Lecture 02 (PDF)

  Plan
- Recall previous lecture. We discussed:
  - Two definitions of lattices: a topological definition and an algebraic/geometric definition
  - Discussed differences between lattices and vectors spaces: in particular, Gram-Schmidt works for vector spaces, but not lattices
- Plan for today's lecture:
  - Characterise lattice bases using unimodular matrices
  - Learn first lattice property: the determinant
- **Characterising Lattice Bases**
  - Recall from last lecture that we saw that some lattices don't have orthogonal basis and we saw that, like vector space bases, lattice bases are not unique. Let's try to characterise lattices bases. To this end, we need to first define unimodular matrices.
  - **Definition 1.** A (square) matrix $\bar{U}$ is unimodular if it has *integer* entries and $det(\bar{U}) = \pm 1$.
    - Intuitively, a unimodular matrix preserves "volume" of the fundamental parallelepiped.
  - **Example 1.** The following are examples of unimodular matrices:
    - Permutation matrices (determinant 1), obtained by swap columns of the identity matrix:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

    - Reflection matrix (determinant -1), obtained by multiplying columns of identity matrix with -1:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

- Certain shear matrices (determinant 1):

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \pm s \\ 0 & 0 & 1 \end{pmatrix}$$

- **HW1.** Show that any unimodular matrix can be obtained by applying the following three column operations to the identity matrix:
  1. Permute: swap two columns
  2. Reflect: multiply a column by -1
  3. Shear: add an integer multiple of one column to another column
- **Theorem 1.** ([2], Theorem 2) Let $\bar{B}_1, \bar{B}_2 \in \mathbb{R}^{n \times m}$ be two bases. Then $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$ iff there exists a unimodular matrix $\bar{U} \in \mathbb{Z}^{m \times m}$ such that $\bar{B}_2 = \bar{B}_1 \bar{U}$.
  - **PQ1.** Given Theorem 1, how do you enumerate all bases for a lattice Λ?
  - **Proof of Theorem 1.**
    - We first prove the following lemma about unimodular matrices:
    - **Lemma 1.** If $\bar{U} \in \mathbb{Z}^{m \times m}$ is a unimodular matrix, then so is $\bar{U}^{-1}$
    - **Proof.** We split the proof into two parts:
      1. In the first part, we argue that $\bar{B} := \bar{U}^{-1}$ is an *integer* matrix. To see why, parse $\bar{V}$ as $(\bar{v}_1, \cdots, \bar{v}_m)$ and each $\bar{v}_i$ as $(v_{i,1}, \cdots, v_{i,m})^t$ and consider the equation

$$\bar{U}\bar{V} = \bar{I}_m,$$

where $\bar{I}_m$ denotes the m-dimensional identity matrix. Using Cramer's rule, the system of linear equations

$$\bar{U}\bar{v}_i = \bar{e}_i,$$

where $\bar{e}_i$ is the ith unit vector, has a unique solution $\bar{v}_i$ where

$$v_{i,j} := \frac{det((\bar{u}_1, \cdots, \bar{u}_{j-1}, \bar{e}_i, \bar{u}_{j+1}, \cdots, \bar{u}_n))}{det(\bar{U})}.$$

Since all the vectors in the numerator are integer, and $det(\bar{U}) = \pm 1$, we have that $\bar{v}_i$ is also an integer vector and thus $\bar{V} := \bar{U}^{-1}$ is also an integer vector.

      2. In the second part, we argue that $det(\bar{U}^{-1}) = \pm 1$. However, this follows by the fact that determinant is multiplicative (see HW2 below):

$$det(\bar{U}) \cdot det(\bar{U}^{-1}) = det(\bar{I}_m) = 1.$$

      - **HW2 (Determinant is multiplicative.)** For any $\bar{M}_1, \bar{M}_2 \in \mathbb{Z}^{m \times m}$, show that

$$det(\bar{M}_1\bar{M}_2) = det(\bar{M}_1)det(\bar{M}_2).$$

- Given Lemma 1, let's prove Theorem 1:
  - $\Leftarrow$ Suppose $\bar{B}_2 = \bar{B}_1\bar{U}$ for some unimodular matrix $\bar{U}$. This implies $\bar{B}_2 \subseteq \Lambda(\bar{B}_1)$ and thus $\Lambda(\bar{B}_2) \subseteq \Lambda(\bar{B}_1)$. Conversely, since $\bar{B}_1 = \bar{B}_2\bar{U}^{-1}$ and since $\bar{U}^{-1}$ is an integer matrix by Lemma 1, we also have $\bar{B}_1 \subseteq \Lambda(\bar{B}_2)$ and thus $\Lambda(\bar{B}_1) \subseteq \Lambda(\bar{B}_2)$. Taken together, $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$.
  - $\Rightarrow$ Suppose $\bar{B}_1 := (\bar{b}_{1,1}, \cdots, \bar{b}_{1,m}), \bar{B}_2 := (\bar{b}_{2,1}, \cdots, \bar{b}_{2,m}) \in \mathbb{R}^{n\times m}$ be two bases such that $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$. This implies, in particular, that $\bar{b}_{1,1}, \cdots, \bar{b}_{1,m} \in \Lambda(\bar{B}_2)$ and vice versa. Thus, by definition of lattices, there exists two *integer* matrices $\bar{M}_1, \bar{M}_2 \in \mathbb{Z}^{m\times m}$ such that $\bar{B}_2 = \bar{B}_1\bar{M}_1$ and $\bar{B}_1 = \bar{B}_2\bar{M}_2$. Substituting the former into the latter, we have

    $$\bar{B}_1 = \bar{B}_1\bar{M}_1\bar{M}_2$$

    and thus

    $$\bar{B}_1(\bar{I}_m - \bar{M}_1\bar{M}_2) = \bar{0}_{m\times m},$$

    where $\bar{0}_{m\times m}$ is the $m \times m$ matrix with zero entries. Since $\bar{B}_1$ is a basis and its column vectors are linearly independent, it follows that $\bar{M}_1\bar{M}_2 = \bar{I}_m$ and thus

    $$det(\bar{M}_1\bar{M}_2) = 1.$$

    Since $\bar{M}_1$ and $\bar{M}_2$ are both integer matrices, their determinants are also integers and thus only way the above equation is satisfied (by HW2) is when $det(\bar{M}_1) = det(\bar{M}_2) = \pm 1$.
- **HW3 (Determinant is basis-independent).** Show that if $\bar{B}_1$ and $\bar{B}_2$ are bases of the same lattice, then $det(\Lambda(\bar{B}_1)) = det(\Lambda(\bar{B}_2)$.
  - Hint: use Theorem 1.
- **HW4 (Existence vs search).** Theorem 1 proves the *existence* of a unimodular matrix $\bar{U}$. Can you find the $\bar{U}$ efficiently?

- **Lattice Determinant.** In this section we discuss our first fundamental lattice property: the *determinant*. Geometrically, it captures *density* of lattice points in its ambient space: if the determinant is small then the lattice is dense and vice versa. Before defining the determinant, we define the fundamental parallelepiped (or region) of a lattice
  - **Definition 2.** (Fundamental parallelepiped) Given a basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_m) \in \mathbb{R}^{n\times m}$, the *fundamental parallelepiped* associated with $\bar{B}$ is the region

    $$P(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i\bar{b}_i : 0 \le x_i < 1 \right\}$$

  - Geometrically, when $m = n$, the whole space $\mathbb{R}^n$ can be obtained by "tiling" using $P(\bar{B})$.

- **Definition 3.** (Determinant) Let $\bar{B} \in \mathbb{R}^{n \times m}$ be a basis. The determinant of $\Lambda(\bar{B})$ is the m-dimensional volume of $P(\bar{B})$:

$$det(\Lambda(\bar{B})) = vol(P(\bar{B})).$$

  - Note that when $m = n$, the definition is the same as that of matrix determinant.
- The following homework asks you to prove a loose upper bound on the determinant. Then you will show that the determinant can be exactly computed using the Gram-Schmidt algorithm, which we discussed Lecture 02, and discussed to be efficient.
  - **HW4 (Hadamard inequality).** Show that $det(\Lambda(\bar{B})) \leq \Pi_{i=1}^{m} ||\bar{b}_i||$.
  - **HW5 (Determinant is efficiently computable).** Let $\bar{B}^* := (\bar{b}_1^*, \cdots, \bar{b}_m^*) \in \mathbb{R}^{n \times m}$ be the basis obtained by running the Gram-Schmidt algorithm on $\bar{B}$ in Definition 5. Then

$$det(\Lambda(\bar{B})) = \Pi_{i=1}^{m} ||\bar{b}_i^*||.$$

- We'll now see how to reduce computing the lattice determinant to matrix determinant, which is much more efficient (and stable) than Gram-Schmidt.
- **Theorem 2 (Theorem 6, in [2]).** For any lattice basis $\bar{B} \in \mathbb{R}^{n \times m}$,

$$det(\Lambda(\bar{B}) = \sqrt{det(\bar{B}^t \bar{B})}.$$

- *Proof.* First observe that the Gram-Schmidt algorithm is captured by the matrix relation

$$\bar{B} = \bar{B}^* \bar{T},$$

  where $\bar{T} \in \mathbb{R}^{m \times m}$ is the upper triangular matrix with 1s on the diagonal and for all $1 < j < i \leq m$, the value

$$\mu_{i,j} := \frac{\langle \bar{b}_2, \bar{b}_1^* \rangle}{\langle \bar{b}_1^*, \bar{b}_1^* \rangle}$$

  in cell (j, i). For example, for $m = 4$,

$$\bar{T} := \begin{pmatrix} 1 & \mu_{2,1} & \mu_{3,1} & \mu_{4,1} \\ 0 & 1 & \mu_{3,2} & \mu_{4,2} \\ 0 & 0 & 1 & \mu_{4,3} \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

If follows that

$$\begin{aligned}\sqrt{det(\bar{B}^t\bar{B})} &= \sqrt{det((\bar{B}^*\bar{T})^t\bar{B}^*\bar{T})} \\ &= \sqrt{det(\bar{T}^t{\bar{B}^*}^t\bar{B}^*\bar{T})} \\ &= \sqrt{det({\bar{B}^*}^t\bar{B}^*)} \\ &= \sqrt{\Pi_i\langle\bar{b}_i^*,\bar{b}_i^*\rangle} \\ &= \sqrt{\|\bar{b}_i^*\|^2} \\ &= det(\Lambda(\bar{B}))\end{aligned}$$

- In the following homework, you'll show that every integer lattice contains an orthogonal sub-lattice:
- **HW6.** For any non-singular $\bar{B} \in \mathbb{Z}^{m\times m}$, $det(\bar{B})\mathbb{Z}^n \subseteq \Lambda(\bar{B})$.
    - Hint: use Cramer's rule.

2026s-CS758-Lecture04

Topics

- Minimum distance
- Blichtfeldt's and Minkowski's theorems

Sources

1. CS206A, Lecture 03 (COURSE-Micciancio07)
2. ToTD article
3. Paper 1: PPP-Completeness with Connections to Cryptography (FOCS-SotZamZir18 STUB)
4. Paper 2: Reducitons in PPP (IPL-BJPPR19 STUB)

Plan

- In the previous lecture
  - We characterised lattice bases using *unimodular matrices*
  - We also defined the first fundamental parameter associated with a lattice: its *determinant*
- In today's lecture
  - We will encounter another fundamental parameter in the form of *minimum distance*
    - Interesting from coding theory and cryptography point of view
  - We will estimate the minimum distance from below and above, and in this process state and prove two interesting theorems about lattices: Blichtfeldt's theorem and Minkowski's theorem
  - Finally, we will use Minkowski's theorem to prove Fermat's theorem on sum of two squares
- Minimum distance
  - Two definitions, equivalent because lattice is a subgroup of $(\mathbb{R}, +)$
    - **Definition 1.** The minimum distance of a lattice Λ is defined as

$$\lambda_1(\Lambda) := \inf_{\bar{y}_1, \bar{y}_2 \in \Lambda, \bar{y}_1 \neq \bar{y}_2} \|\bar{y}_1 - \bar{y}_2\|$$

    - **Definition 2.** The minimum distance of a lattice Λ is defined as

$$\lambda_1(\Lambda) := \inf_{0^n \neq \bar{y} \in \Lambda} \|\bar{y}\|$$

    - **Example 1 (from [2]).** For the lattice Λ generated by $(2, 0)^t$ and $(1, 1)^t$, $\lambda_1(\Lambda) = \sqrt{2}$.
    - **HW1.** Show that Definition 1 is equivalent to Definition 2.
  - Motivation:

- Cryptography: cryptography can be based on the hardness of *estimating* minimum distance (GapSVP) *in the worst case*!
- Coding theory: (random) lattices are good (linear) error-correcting codes:
  - Lattice basis $\bar{B} \in \mathbb{Z}^{n \times m}$ (recall that $m < n$) corresponds to the generator matrix, and thus the message space consists vectors in $\mathbb{Z}^m$ and the lattice $\Lambda(\bar{B})$ generated by $\bar{B}$ are the codewords.
  - Minimum distance corresponds to the minimum Hamming distance of the code, which corresponds to the unique decoding radius
    - That is, if the corrupted codeword $\bar{y}'$ is in a ball of radius λ_1 of some some codeword (i.e, a lattice point), then unique decoding is possible
  - Therefore, it is desirable to have a *large* minimum distance

- Lower Bound
  - **Claim 1 ([1], Proposition 2).** For a lattice $\Lambda(\bar{B})$, let $\bar{B}^* := (\bar{b}_1^*, \cdots, \bar{b}_m^*)$ denote the Gram-Schmidt orthogonalisation of $\bar{B}$. Then

$$\lambda_1(\Lambda(\bar{B})) \geq \min_{1 \leq i \leq m} \|\bar{b}_i^*\|$$

  - *Proof.*
    - Let $\bar{B}\bar{x}$, $\bar{x} \in \mathbb{Z}^m \setminus \{0^n\}$, be any non-zero lattice vector and let $1 \leq I \leq m$ be the largest index such that $x_I \neq 0$ (i.e., $\bar{x}$ is of the form $(x_1, \cdots, x_I, 0, \cdots, 0)^t$). We first show that

$$\|\bar{B}\bar{x}\| \geq \|\bar{b}_I^*\| \geq \min_{1 \leq i \leq m} \|\bar{b}_i^*\|.$$

      To see why observe that

$$\begin{aligned} \langle \bar{B}\bar{x}, \bar{b}_I^* \rangle &= \sum_{1 \leq i \leq I} \langle x_i \bar{b}_i, \bar{b}_I^* \rangle \\ &= x_I \langle \bar{b}_I, \bar{b}_I^* \rangle \\ &= x_I \|\bar{b}_I^*\|^2, \end{aligned}$$

      where the second and third equalities follow by orthogonality of $\bar{b}_I^*$ and $\bar{b}_i$s for $i < I$. Now, by Cauchy-Schwartz inequality and since $|x_I| \geq 1$, we get

$$\|\bar{B}\bar{x}\| \cdot \|\bar{b}_I^*\| \geq |\langle \bar{B}\bar{x}, \bar{b}_k^* \rangle| = |x_I| \cdot \|\bar{b}_I^*\|^2 \geq \|b_I^*\|^2$$

      and thus

$$\|\bar{B}\bar{x}\| \geq \|\bar{b}_I^*\| \geq \min_{1 \leq i \leq m} \|\bar{b}_i^*\|.$$

  - Using Claim 1, it can be shown that there exists a lattice point $\bar{y}$ such that $\|\bar{y}\| = \lambda_1$:
    - **HW2.** Show that the infimum in Definitions 1 and 2 can be replaced by minimum.
- Upper Bound

- To upper bound λ_1, we prove the existence of a lattice point of certain length determined by its determinant (and dimension) and hence independent of the basis
- **Theorem 1 ([1], Corollary 6).** For any full-rank lattice $\Lambda(\bar{B})$, there exists a non-zero lattice point $\bar{y}$ such that

$$\|\bar{y}\| \leq \sqrt{n} \cdot det(\bar{B})^{1/n}$$

- Interestingly, Theorem 1 boils down to the pigeonhole principle! We will prove the claim in three steps:
  1. First, we will prove Blichtfeldt's theorem, which informally says that any large-enough region $\mathcal{S}$ of the ambient space must have two (not-necessarilty-lattice) points such that their difference is a lattice point:
     - **Theorem 2 ([1], Theorem 3).** For a full-rank lattice $\Lambda(\bar{B})$ and a set $\mathcal{S} \subseteq \mathbb{R}^n$, if

       $$vol(\mathcal{S}) > vol(P(\bar{B})) =: det(\bar{B}),$$

       then $\mathcal{S}$ contains two points $\bar{z}_1 \neq \bar{z}_2$ such that $\bar{z}_1 - \bar{z}_2 \in \Lambda(\bar{B})$.
     - **Example 2.** Consider Λ from Example 1, whose fundamental parallelepiped has volume 2 (units)
     - Proof
       1. Let's *partition* $\mathcal{S}$ using *translated copies* of the fundamental paralellepiped $P(\bar{B})$. That is,

          $$\mathcal{S} = \cup_{\bar{y} \in \Lambda(\bar{B})} \mathcal{S}_{\bar{y}},$$

          where for $\bar{y} \in \Lambda(\bar{B})$ we define

          $$\mathcal{S}_{\bar{y}} := \mathcal{S} \cap (\bar{y} + P(\bar{B})).$$

          Note that

          $$vol(\mathcal{S}) = \sum_{\bar{y} \in \Lambda(\bar{B})} vol(\mathcal{S}_{\bar{y}}).$$

       2. Next, consider these partitions *translated* to the fundamental parallelepiped, i.e., the sets

          $$\mathcal{S}_{\bar{y}} - \bar{y} = (\mathcal{S} - \bar{y}) \cap P(\bar{B}) \subseteq P(\bar{B}).$$

       3. By pigeonhole principle the translated partitions $\mathcal{S}_{\bar{y}} - \bar{y}$ cannot be mutually disjoint since

          $$\sum_{\bar{y} \in \Lambda(\bar{B})} vol(\mathcal{S}_{\bar{y}} - \bar{y}) = \sum_{\bar{y} \in \Lambda(\bar{B})} vol(\mathcal{S}_{\bar{y}}) = vol(\mathcal{S}) > vol(P(\bar{B})).$$

Therefore there exist two vectors $\bar{y}_1 \neq \bar{y}_2 \in \Lambda(\bar{B})$ such that

$$(\mathcal{S}_{\bar{y}_1} - \bar{y}_1) \cap (\mathcal{S}_{\bar{y}_2} - \bar{y}_2) \neq \emptyset.$$

4. Let's call such an (non-lattice) "collision" vector $\bar{z}$ and define

$$\bar{z}_1 := \bar{z} + \bar{y}_1 \in \mathcal{S}_{\bar{y}_1}, \bar{z}_2 := \bar{z} + \bar{y}_2 \in \mathcal{S}_{\bar{y}_2}$$

to be their "pre-image" in the original partitions. Now note that

$$\bar{z}_1 - \bar{z}_2 = \bar{y}_1 - \bar{y}_2 \in \Lambda(\bar{B}).$$

2. Second, we will use Blichtfeldt's theorem to prove Minkowski convex-body theorem
   - **Theorem 3 ([1], Corollary 4).** Consider a lattice $\Lambda(\bar{B})$. If $\mathcal{C} \subseteq \mathbb{R}^n$ is a *convex*, *symmetric* set such that

     $$vol(\mathcal{C}) > 2^n det(\bar{B}),$$

     then $\mathcal{C}$ contains a (non-zero) *lattice point*. Recall that:
     - If $\mathcal{C}$ is *convex* then for any $\bar{z}_1, \bar{z}_2 \in \mathcal{C}$, $(\bar{z}_1 + \bar{z}_2)/2 \in \mathcal{C}$.
     - $\mathcal{C}$ is symmetric if $\bar{z} \in \mathcal{C} \Rightarrow -\bar{z} \in \mathcal{C}$
   - *Proof.* Consider the set

     $$\mathcal{C}/2 := \{\bar{z} : 2\bar{z} \in \mathcal{C}\},$$

     and note that

     $$vol(\mathcal{C}/2) = vol(\mathcal{C})/2^n > det(\bar{B}).$$

     Applying Theorem 1 with $\mathcal{S} := \mathcal{C}/2$, there exist $\bar{z}_1 \neq \bar{z}_2 \in \mathcal{C}/2$ such that

     $$\bar{y} := \bar{z}_1 - \bar{z}_2 \in \Lambda(\bar{B}) \setminus \{0^n\}.$$

     We now claim that $\bar{y}$ is also in $\mathcal{C}$ by symmetry and convexity:
     1. By definition of $\mathcal{C}/2$, $2\bar{z}_1, 2\bar{z}_2 \in \mathcal{C}$.
     2. By symmetry $-2\bar{z}_2 \in \mathcal{C}$.
     3. By convexity

     $$\mathcal{C} \ni (2\bar{z}_1 - 2\bar{z}_2)/2 = \bar{z}_1 - \bar{z}_2 = \bar{y}.$$

3. Finally, we will use Minkowski's theorem to prove Theorem 1.
   - **HW3.** Using Theorem 3, show that for any full-rank lattice $\Lambda(\bar{B})$, there exists non-zero lattice point $\bar{y}$ such that

     $$\|\bar{y}\|_\infty \leq det(\bar{B})^{1/n},$$

     where $\|\cdot\|_\infty$ denotes the ℓ_∞ norm, i.e.,

     $$\ell_\infty(\bar{y}) := \max_{1 \leq i \leq n} |y_i|.$$

        - **HW4.** Show that $\|\bar{y}\| \leq \sqrt{n} \cdot \|\bar{y}\|_\infty$.
    - Estimates on λ_1 via Minkowski's theorem need not be tight
        - **Example 1.** Consider the 2D lattice generated by $(1,0)^t$ and $(0,N)^t$ for a large $N \in \mathbb{N}$. The lattice contains a short vector $(1,0)^t$ of length 1, but its determinant is N and Minkowski's bound $\sqrt{2N}$ is much larger than 1.
    - Conversely, there are lattices where Minkowski's bound is close to tight
        - **HW4.** For any n, there is an n dimensional lattice Λ_n such that

$$\lambda_1(\Lambda_n) \geq c\sqrt{n} \cdot det(\Lambda_n)^{1/n}.$$

    - Computational problems associated with Theorems 2 and 3:
        - Theorems 2 and 3 are existence theorems, and have natural search counterparts [3,4]:
            - MINKOWSKI
                - Inputs: full-rank lattice $\Lambda(\bar{B})$ of dimension n of determinant at least $\sqrt{n} \cdot 2^n$
                - Solution: non-zero lattice point in $[-1,1]^n$
            - BLICHTFELDT
                - Inputs: full-rank lattice $\Lambda(\bar{B})$ of dimension n and $\mathcal{S} \subseteq \mathbb{R}^n$ such that $vol(\mathcal{S}) > vol(P(\bar{B}))$ described using a succinct a circuit
                - Solution: $\bar{y}_1 \neq \bar{y}_2 \in \mathcal{S}$ such that $\bar{y}_1 - \bar{y}_2 \in \Lambda(\bar{B})$
        - Known to be in a complexity class called $\mathbf{PPP}$, which contain problems "reducible to " pigeonhole principle
    - **Open Question 1.** Are MINKOWSKI and BLICHTFELDT *complete* for $\mathbf{PPP}$?
- Application
    - **Theorem 3 ([1], Theorem 7).** For every prime $p \equiv 1 \bmod 4$, there exists integers $a, b \in \mathbb{Z}$ such that $p = a^2 + b^2$.
    - *Proof.* See [1]

2026s-CS758-Lecture05

Topics

- Successive minima
- Shortest-vector problem (SVP) and its friends
- Lagrange's algorithm for 2D-SVP

Sources

1. Intro to Lattice Algorithms and Cryptography, Lecture 2
2. Intro to Lattice Algorithms and Cryptography, Lecture 3
3. Intro to Lattice Algorithms and Cryptography, Lecture 4
4. The LLL Algorithm: Survey and Algorithms, Chapter 02

Plan

- Plan/Recall
  - In the last lecture,
    - We defined the minimum distance $\lambda_1(\Lambda)$ of a lattice Λ and saw how it is relevant to cryptography and coding theory
    - Then we went on to establish upper and lower bounds for the minimum distance. In particular, we saw Minkowski's first theorem, which upper bounds λ_1 in terms of the determinant (which is basis-independent):

$$\lambda_1(\Lambda(\bar{B})) \leq \sqrt{n} \cdot det(\bar{B})^{1/n}$$

      - Essentially a consequence of pigeonhole principle via Blichtfeldt's theorem and Minkowski's convex-body theorem
  - In today's lecture:
    - We first define *successive minima*, which generalises the notion of minimum distance
    - Then we will introduce computational problems related to minimum distance: the shortest vector problem (SVP) and its approximate variant
    - In general SVP is a hard problem, we'll see that for the 2D case, there is an efficient algorithm: the Lagrange's algorithm. Along the way, we will learn reduced bases
- Successive Minima
  - Recall the definition of λ_1: it can be seen as the radius of the smallest ball that contains a non-zero lattice point
  - Informally, λ_i is the radius of the smallest ball that contains i *linearly-independent* lattice points

- **Example 1.** Consider $\Lambda := \Lambda((1,0)^t, (3,4)^T)$. Note that $\lambda_1(\Lambda) = 1$ since $(1,0)^t \in \Lambda$. Note that $(2,0)^t$ is also in Λ, but it is linearly dependent on $(1,0)^t$. It seems that $\lambda_2(\Lambda) = 4$ since $(0,4) \in \Lambda$.
- More formally:
  - **Definition 1.** For $1 \leq i \leq m$, we define

    $$\lambda_i(\Lambda) := \inf_{r \geq 0} dim(\Lambda \cap B_r^n) \geq i,$$

    where B_r^n is the n-dimensional ball of radius r
    - Note that $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_m$
  - **HW1.**
    - Let $\Lambda \subset \mathbb{R}^n$ be a m-rank lattice. Show that there exists vectors $\bar{y}_1, \cdots, \bar{y}_m \in \Lambda$ such that $\|\bar{y}_i\| = \lambda_i$.
    - Does Λ have a basis $\bar{b}_1 \cdots, \bar{b}_m$ such that $\|\bar{b}_i\| = \lambda_i$? This is true for Λ in Example 1, but is it true in general?
- Minkowski also proved the following theorem bounding the *geometric mean* of successive minima:
  - **Theorem 1 (Minkowski's second theorem, see [1], Theorem 21).** For any full-rank lattice Λ of dimension n,

    $$(\Pi_{i=1}^n \lambda_i(\Lambda))^{1/n} \leq \sqrt{n} \cdot det(\Lambda)^{1/n}$$

    - Note that Theorem 2 is a stronger bound than Minkowski's first theorem since the mean is at least λ_1.

- Shortest-Vector Problem and its Friends
  - We are now ready to define some computational problems on lattices
    - **Definition 2 (SVP: Shortest Vector Problem, [3], Definition 1).**
      - *Input.* A basis $\bar{B}$ of lattice Λ
      - *Solution.* A vector $\bar{y} \in \Lambda$ such that $\|\bar{y}\| = \lambda_1(\Lambda)$
    - **Definition 3 (SVP$_\alpha$: Approximate SVP, [3], Definition 1).**
      - *Input.* A basis $\bar{B}$ of a lattice Λ of dimension n and an approximation factor $\alpha(\cdot)$
      - *Solution.* A vector $y \in \Lambda$ such that $\|\bar{y}\| \leq \alpha(n)\lambda_1(\Lambda)$
    - **Definition 4 (GapSVP$_\gamma$: gap SVP, [4], Definition 1).**
      - *Input.* A basis $\bar{B}$ of lattice Λ, a distance $d \in \mathbb{Q}$ and an gap factor $\gamma(n)$
      - *Decision.* Is $\lambda_1(\Lambda) > \gamma(n)d$ or $\lambda_1(\Lambda) \leq d$?
    - **Definition 5 (SIVP$_\alpha$: Approximate Shortest Independent Vector Problem, [4], Definition 2).**
      - *Input.* A basis $\bar{B}$ of lattice Λ and an approximation factor $\alpha(\cdot)$
      - *Solution.* m linearly-independent lattice vectors $\bar{y}_1, \cdots, \bar{y}_m$ such that $\|\bar{y}_i\| \leq \alpha(n)\lambda_n(\Lambda)$
  - What do we know about these problems? (Source)

- SVP
  - Ajtai showed that SVP is NP-hard using a decision-to-search (randomised) reduction from *any* NP problem to SVP
    - Priyanshu will cover basics of complexity theory in Recitation 2.
  - We don't know how to base cryptography on SVP
- SVP$_\alpha$
  - Khot showed that SVP is NP-hard for any constant approximation factor, i.e, $\alpha := O(1)$
  - For approximation factors $\alpha(n) := \sqrt{n/\log(n)}$ and $\alpha(n) := O(\sqrt{n})$, SVP$_\alpha$ was shown to be in $\mathbf{coNP}$ and $\mathbf{coAM}$ respectively by Aharonov and Regev and Goldreich and Goldwasser
  - For approximation factors $\alpha(n) := n$, it seems we can do cryptography (I was wrong!).
  - In the next lecture, we will see the LLL algorithm, which solves SVP$_\alpha$ for $\alpha := 2^{O(n)}$.
- Here is a very nice schematic by Vinod Vaikuntanathan capturing all of the above:

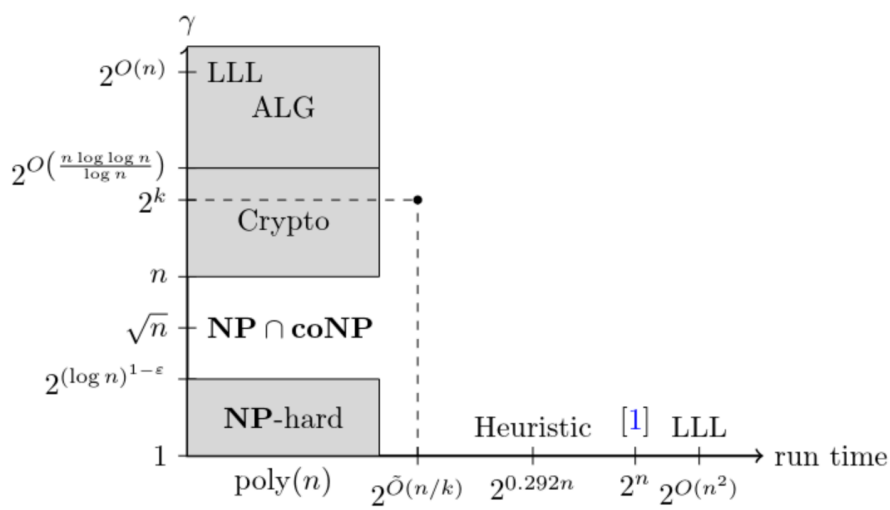

Figure 1: The complexity landscape of SVP$_\gamma$.

- GapSVP$_\gamma$
  - We can build cryptography from GapSVP$_\gamma$ for $\gamma(n) := O(n)$: see the work of Micciancio and Regev
- SIVP$_\alpha$
  - Ajtai showed how to base cryptography on approximate SIVP for $\alpha := poly(n)$. We will see this result in Module II

- Lagrange's algorithm for 2D-SVP
  - In this section we will see that although (exact) SVP is $\mathbf{NP}$-hard, the hardness only really kicks in in higher dimensions. That is, we will see a polynomial-time algorithm for SVP when $n = 2$, called Lagrange's algorithm. In this process we will

introduce the notion of a "reduced" basis. The techniques used will be useful in the next lecture when we discuss the celebrated LLL algorithm for *approximate* SVP.

- *Reduced basis.* Informally a basis is "reduced" if consists of short, almost orthogonal vectors. There are several notions of reduced. In this lecture, we will consider "Lagrange" reduced basis for $\mathbb{R}^2$, while in the next lecture we will consider "LLL-reduced" basis for $\mathbb{R}^n$.
  - **Definition (Wristwatch Condition).** A basis $(\bar{b}_1, \bar{b}_2)$ for a lattice $\Lambda \subset \mathbb{R}^2$ is Lagrange-reduced iff

    $$\|\bar{b}_1\| \leq \|\bar{b}_2\|$$

    and

    $$|\langle \bar{b}_1, \bar{b}_2 \rangle| \leq \|\bar{b}_1\|^2/2.$$

    - Geometrically, it looks like a wrist-watch: see figure.
  - In the following claim, we show that a Lagrange-reduced basis is in fact the "shortest basis"!
  - **Claim 1.** For a Lagrange-reduced basis $(\bar{b}_1, \bar{b}_2)$, $\|\bar{b}_1\| = \lambda_1(\Lambda)$ and $\|\bar{b}_2\| = \lambda_2(\Lambda)$.
  - *Proof.*
    1. First we show that $\|\bar{y}\| \geq \|\bar{b}_1\|$ for any non-zero $\bar{y} \in \Lambda$. Since $\bar{y} \in \Lambda$, there exists $x_1, x_2 \in \mathbb{Z}$ such that $\bar{y} = x_1\bar{b}_1 + x_2\bar{b}_2$. Now we have,

       $$\begin{aligned} \|\bar{y}\|^2 &= \|x_1\bar{b}_1 + x_2\bar{b}_2\|^2 \\ &= x_1^2\|\bar{b}_1\|^2 + 2x_1x_2\langle \bar{b}_1, \bar{b}_2 \rangle + x_2^2\|\bar{b}_2^2\| \\ &\geq x_1^2\|\bar{b}_1^2\| - x_1x_2\|\bar{b}_1\|^2 + \|\bar{b}_2^2\| \\ &\geq (x_1^2 - x_1x_2 + x_2^2)\|\bar{b}_1\|^2 \\ &\geq \|\bar{b}_1\|^2, \end{aligned}$$

       where
       - The first inequality holds by the second clause of wristwatch condition, i.e., $\langle \bar{b}_1, \bar{b}_2 \rangle \geq -\|\bar{b}_1\|^2/2$,
       - The second inequality holds by its first clause, and
       - The last inequality holds since $\bar{y}$ is non-zero and both $x_1 = 0$ and $x_2 = 0$ cannot hold
    2. Next, we show that $\|\bar{y}\| \geq \|\bar{b}_2\|$ whenever $x_2 \neq 0$ (i.e., $\bar{y}$ is not linearly dependent on $\bar{b}_1$.
       - **HW2.** Prove the second part of Claim 1.
- *The algorithm.*
  - The algorithm is similar to Euclid's GCD algorithm, which we now recall:
    - **Algorithm 1**:
      - *Input*: $a, b \in \mathbb{Z}$
      - *Output*: $gcd(a, b)$
      - Algorithm:

1. If $|a| \leq |b|$ then swap a and b
2. Else, while $b \neq 0$ do:
    1. Compute $r := a - \lfloor a/b \rceil b$, where $\lfloor \cdot \rceil$ is the function that rounds to the nearest integer (say breaking ties upwards)
    2. Set $a := b$ and $b := r$
3. Output a

- **Example 2.** Running Algorithm 1 on $(56, 315)$, results in the sequence $(315, 56)$, $(56, 35)$, $(35, 21)$, $(21, 14)$, $(14, 7)$, $(7, 0)$

- The idea is to carry out Gram-Schmidt *with rounding* and swap the resulting basis vectors as long as $\bar{b}_1$'s length is greater than $\bar{b}_2$
    - **Algorithm 2**:
        - *Input*: Basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$
        - *Output*: Lagrange-reduced basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$
        - Algorithm:
            1. If $\|\bar{b}_1\| < \|\bar{b}_2\|$ then swap $\bar{b}_1$ and $\bar{b}_2$
            2. Else, repeat
                1. Compute $\bar{r} := \bar{b}_1 - \lfloor \mu_{1,2} \rceil \bar{b}_2$, where if you recall $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \langle \bar{b}_2, \bar{b}_2 \rangle$
                2. Set $\bar{b}_1 := \bar{b}_2$ and $\bar{b}_2 := \bar{r}$.
                Until $\|\bar{b}_1\| \leq \|\bar{b}_2\|$.
            3. Output $(\bar{b}_1, \bar{b}_2)$
    - **Example 3.** On running Algorithm 2 on $((1,5)^t, (6,21)^t)$ results in the sequence $((6,21)^t, (1,5)^t)$, $((1,5)^t, (2,1)^t)$ and $(2,1)^t, (-1,4)^t)$. See figure
- **Theorem 2.** Algorithm 2 runs for $O(\log(\|\bar{b}_1\| / \sqrt{det(\Lambda)})$ iterations and outputs a Lagrange-reduced basis.
- *Proof.*
    - **Claim 2.** Algorithm 2 outputs a Lagrange-reduced basis.
    - *Proof.*
        - **Pop-Quiz 1.** Why does the output forms a basis of Λ? Because we are only doing the column operations corresponding to unimodular matrix
        - **Pop-Quiz 2.** Why does the wristwatch condition hold?
            - First, note that Algorithm 1 *terminates* since $\|\bar{b}_1\|$ diminishes in each iteration: we are replacing $\bar{b}_1$ with the shorter of the basis vectors, i.e., $\bar{b}_2$, and $\bar{b}_2$ is shrinking since we are replacing it with "rounded" orthogonal projections of $\bar{b}_1$ (similar to Gram-Schmidt). Since termination can only happen when $\|\bar{b}_1\| \leq \|\bar{b}_2\|$, the first wristwatch condition holds.

- To see why, let's denote the basis at the end of each iteration by $(\bar{b}_1', \bar{b}_2')$ observe by construction that

$$\begin{aligned}\langle \bar{b}_1', \bar{b}_2' \rangle &= \langle \bar{b}_2, \bar{b}_1 - \lfloor \mu_{1,2} \rceil \bar{b}_2 \rangle \\ &= \langle \bar{b}_1, \bar{b}_2 \rangle - \lfloor \mu_{1,2} \rceil \|\bar{b}_2\|^2\end{aligned}$$

  Showing that the above expression is at most $\|\bar{b}_1'\|^2/2 = \|\bar{b}_2\|^2/2$ is the same as showing

$$\langle \bar{b}_1, \bar{b}_2 \rangle / \|\bar{b}_2\|^2 - \lfloor \mu_{1,2} \rceil \leq 1/2.$$

  However, this is true since $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \|\bar{b}_2\|^2$ and definition of rounding.

- **Claim 3.** Algorithm 2 runs for $O(\log(\|\bar{b}_1\| / \sqrt{det(\Lambda)})$ iterations.
  - **HW3.** Show that the length of $\bar{b}_1$ reduces by a constant factor in each iteration. What exactly is the constant

2026s-CS758-Lecture06

Topics

- Hermite's bound and reduction algorithm
- Weak LLL algorithm

Sources

1. Intro to Lattice Algorithms and Cryptography, Lecture 3
2. Intro to Lattice Algorithms and Cryptography, Lecture 4
3. The LLL Algorithm: Survey and Applications, Chapter 02 (PDF)

Plan

- Recall/Plan
  - In the last lecture:
    - We saw Lagrange's reduction, a poly-time *exact* algorithm for SVP in 2D. Recall the algorithm:
      - **Algorithm 1**:
        - *Input*: Basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$
        - *Output*: A basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$ with $\|\bar{b}_1\| \leq \|\bar{b}_2\|$ and Gram-Schmidt coefficient $\mu_{1,2} \leq 1/2$ (recall wristwatch figure), which implies $\|\bar{b}_1\| = \lambda_1$
        - Algorithm:
          1. If $\|\bar{b}_1\| < \|\bar{b}_2\|$ then swap $\bar{b}_1$ and $\bar{b}_2$
          2. Else, repeat
             1. Compute $\bar{r} := \bar{b}_1 - \lfloor \mu_{1,2} \rceil \bar{b}_2$, where if you recall $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \langle \bar{b}_2, \bar{b}_2 \rangle$
             2. Set $\bar{b}_1 := \bar{b}_2$ and $\bar{b}_2 := \bar{r}$.
             Until $\|\bar{b}_1\| \leq \|\bar{b}_2\|$.
          3. Output $(\bar{b}_1, \bar{b}_2)$
    - We argued that Algorithm 1 runs in polynomial time since the length of $\bar{b}_1$ reduces by at least half in each iteration.
    - A natural question to ask is what happens when you run Lagrange's reduction in higher dimensions.
  - In today's lecture:
    - We will see Hermite's reduction, a natural generalisation of Lagrange's reduction to higher dimensions that solves *approximate SVP* . The approximation factor turns out to be exponential in the dimension n. To this end we will define Hermite's constant.

    - However, we don't know whether Hermite's reduction runs in polynomial time. We partially resolve this by tweaking Hermite's reduction to obtain a *weak LLL* reduction, in which the number of *iterations* is polynomial in the dimension.
    - In the next lecture we will see how to further tweak Hermite's reduction to obtain the celebrated LLL reduction, which *can* be shown to be poly-*time*.
- Hermite's Bound
    - Recall Minkowski's first theorem, which bounds the minimum distance of a rank-n lattice Λ as

$$\lambda_1(\Lambda) \leq \sqrt{n} \cdot \det(\Lambda)^{1/n}.$$

    - An older, weaker bound on minimum distance, due to Hermite, is stated below in Theorem 1.
    - **Definition 1 (Definitions 7 and 8 in [1]).** For a full-rank lattice $\Lambda \subset \mathbb{R}^n$, we define

$$\gamma_n(\Lambda) := (\lambda_1(\Lambda)/\det(\Lambda)^{1/n})^2.$$

      Hermite's constant is defined as

$$\gamma_n := \sup_{\Lambda \text{ of rank } n} \gamma_n(\Lambda).$$

    - **Theorem 1 (Lemma 9 in [1] and Theorem 5 in [2]).** We have

$$\gamma_2 = \sqrt{4/3}$$

      and

$$\gamma_n \leq \gamma_2^{n-1}.$$

    - A couple of remarks on Theorem 1 are in order:
      1. The supremum for γ_2 is achieved by the hexagonal lattice, generated by $((0,1),(\sqrt{3/4},1/2))$: see Figure 1.
      2. A corollary to Theorem 1 is that for any rank-n lattice Λ

$$\lambda_1(\Lambda) \leq (4/3)^{(n-1)/4} \cdot \det(\Lambda)^{2/n},$$

         which is quantitatively weaker than Minkowski's bound. Nevertheless, the bound is interesting since Hermite's reduction and weak LLL reduction can be regarded as an algorithmisation of its proof, and thus yield an $2^{O(n)}$-*approximate* solution to SVP.
- Hermite's Reduction
    - Before extending Lagrange's algorithm to higher dimensions, we need to define two notions: orthogonal complement (which we already saw in Gram-Schmidt) and lattice projections.

- **Definition 2.** For a vector space V, the *orthogonal complement* of a linear subspace $W \subseteq V$ is defined as

$$W^{\perp} := \{\bar{b} \in V : \langle \bar{v}, \bar{w} \rangle \text{ for all } \bar{w} \in W\}.$$

  If W has a basis $(\bar{w}_1, \cdots, \bar{w}_k)$ then we also write $W^{\perp}$ as $(\bar{w}_1, \cdots, \bar{w}_k)^{\perp}$.
- **Notation 1.** For a basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ and $1 \leq i \leq j \leq n$, we use $\pi_i(\bar{b}_j)$ to denote the *projection* of $\bar{b}_j$ on to the orthogonal complement $(\bar{b}_1, \cdots, \bar{b}_i)^{\perp}$.
  - Note that using this notation, the Gram-Schmidt basis $\bar{B}^*$ of $\bar{B}$ can be denoted by

$$(\pi_1(\bar{b}_1), \pi_2(\bar{b}_2), \cdots, \pi_n(\bar{b}_n))$$

- **Definition 3.** For a lattice Λ generated by $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ and $1 \leq i \leq j \leq n$, we define

$$\bar{B}_{i:j} : (\pi_i(\bar{b}_i), \cdots, \pi_i(\bar{b}_j))$$

  and the projected lattice $\Lambda_{i:j}$ as the lattice generated by $\bar{B}_{i:j}$, i.e.,

$$\Lambda_{i:j} := \Lambda((\pi_i(\bar{b}_i), \cdots, \pi_i(\bar{b}_j))$$

- **Example 1.** For a lattice $\Lambda((1,0,0)^t, (0,1,5)^t, (0,6,21)^t))$ (see Figure 2), $\bar{B}_{2:3}$ is $((0,1,5)^t, (0,6,21)^t)$ since $(0,1,5)^t$ and $(0,6,21)^t$ are already orthogonal to $(1,0,0)$. Hence $\Lambda_{2:3} = \Lambda(((0,1,5)^t, (0,6,21)^t))$.
- **Homework 1.** Prove that

$$\det(\Lambda_{i:j}) = \Pi_{k=i}^{j} \|\bar{b}_k^*\|.$$

- *Hermite's reduction.* We are now ready to state the algorithm, which "locally" carries out Lagrange reduction between pairs of bases in consecutive dimensions till all bases are Lagrange reduced.
- **Algorithm 2**:
  - *Input*: Basis $(\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$
  - *Output*: A basis $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq \gamma_2^{n-1} \cdot \det(\Lambda)^{2/n}$
  - Algorithm:
    1. While $\exists i \in [n]$ such that $\bar{B}_{i:i+1} = (\pi_i(\bar{b}_i), \pi_i(\bar{b}_{i+1}))$ is not Lagrange-reduced, do:
       1. Find unimodular $\bar{U} \in \mathbb{Z}^{2\times 2}$ such that $(\bar{b}'_i, \bar{b}'_{i+1}) := \bar{B}_{i:i+1}\bar{U} = (\pi_i(\bar{b}_i), \pi_i(\bar{b}_{i+1}))\bar{U}$ is Lagrange reduced
       2. Update $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_{i-1}, \bar{b}'_i, \bar{b}'_{i+1}, \bar{b}_{i+2}, \cdots, \bar{b}_n)$
    2. Output $\bar{B}$.
- **Example 2.** Consider $\Lambda := \Lambda((1,0,0)^t, (0,1,5)^t, (0,6,21)^t)$ (see Figure 3).
  - In the first iteration, $i = 2$ is a candidate since

$$\bar{B}_{2:3} := (\pi_2((0,1,5)^t), \pi_2((0,6,21)^t)) = ((0,1,5)^t, (0,6,21)^t)$$

is not Lagrange-reduced. Carrying out Lagrange reduction on $\bar{B}_{2:3}$ yields (as we saw in last lecture)

$$((0,2,1)^t,(0,-1,4)^t).$$

Thus the update step yields

$$\bar{B}=((1,0,0)^t,(0,2,1)^t,(0,-1,4)^t)).$$

- Since $(1,0,0)^t$ and $(0,2,1)^t$ are orthogonal, and $\|(1,0,0)^t\| \leq \|(0,2,1)^t\|$, they are *already* Lagrange-reduced and we're done.

- **Theorem 2.** Algorithm 2 *terminates* and *outputs* $(\bar{b}_1,\cdots,\bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq \gamma_2^{n-1}\cdot\det(\Lambda)^{2/n}$, i.e., an $2^{O(n)}$-approximate solution to SVP.
- *Proof.* We prove termination and correctness, in that order. But first we prove a useful lemma that bounds the output of Lagrange reduction in terms of γ_2.
  - **Lemma 1.** $\bar{B}_{i:i+1} := (\pi_i(\bar{b}_i),\pi_i(\bar{b}_{i+1}))$ is Lagrange-reduced iff

    $$\|\bar{b}_i^*\| \leq \gamma_2\cdot\|\bar{b}_{i+1}^*\|.$$

  - Proof.
    - ($\Rightarrow$) If $\bar{B}_{i:i+1}$ is Lagrange-reduced then $\pi_i(\bar{b}_i)=\bar{b}_i^*$ is "locally" shortest for $\bar{B}_{i,i+1}$, and we have

      $$\|\bar{b}_i^*\|^2 \leq \gamma_2\cdot\det(\bar{B}_{i:i+1}) = \gamma_2\|\bar{b}_i^*\|\|\bar{b}_{i+1}^*\|,$$

      where the inequality follows by definition of γ_2, while the equality is a consequence of Homework 1. Dividing the above by $\|\bar{b}_i^*\|$ proves the if part.
    - ($\Leftarrow$) The iff part can be proved by reversing the above steps.

1. *Termination.* We proceed by induction in dimension n: assuming that Algorithm 2 terminates for lattices of dimension *at most* n, we show that Algorithm 2 terminates for lattices of dimension n. (Note that the base case is the same as Lagrange's algorithm.)
   - First note that since the application of Lagrange reduction is "local", $\bar{b}_1$'s norm doesn't change if its is *not* picked for Lagrange reduction.
   - When $\bar{b}_1$ *is* picked for Lagrange reduction, it can be argued using Lemma 1 that its norm after the reduction is *strictly reduced*. To see why, note that Hermite reduction applies Lagrange reduction to $\bar{B}_{1:2} := (\pi_1(\bar{b}_1)=\bar{b}_1,\pi_1(\bar{b}_2))$ only when the basis is *not* Lagrange reduced. Thus *before* the reduction, by Lemma 1, we have

     $$\|\bar{b}_1\| = \|\bar{b}_1^*\| > \gamma_2\cdot\bar{b}_{i+1}^* \Leftrightarrow \|\bar{b}_1^*\|^2 > \gamma_2\cdot\det(\Lambda_{1:2}).$$

     But *after* the reduction we have

$$\|\bar{b}_1\|^2 \leq \gamma_2 \cdot \det(\Lambda_{1:2}),$$

and thus the norm has decreased.

- Since λ_1 is always achieved (i.e., infimum is minimum), the algorithm must halt at some point. After this point, the algorithm works only over $\Lambda_{2:n}$, and hence terminates by induction hypothesis. Thus Algorithm 2 terminates also for dimension n.

2. *Correctness.* Since Algorithm 2 terminates, it must be the case that each pair $(\pi_i(\bar{b}_i), \pi_i(\bar{b}_{i+1}))$ is Lagrange reduced and thus (by Lemma 1) satisfies

$$\|\bar{b}_i^*\| \leq \gamma_2 \cdot \|\bar{b}_{i+1}^*\|.$$

It follows that

$$\|\bar{b}_1\| = \|\bar{b}_1^*\| \leq \gamma_2^{i-1} \|\bar{b}_i^*\|.$$

Multiplying the above equation for $i = 1, \cdots, n$, we get

$$\|\bar{b}_1\|^n \leq \Pi_{i=1}^n \gamma_2^{i-1} \|\bar{b}_i^*\|^2 = \gamma_2^{n(n-1)/2} \det(\Lambda),$$

where the last equality follows by Homework 1. Correctness now follows by taking nth root on both sides.

- **Homework 2.** Show that correctness of Algorithm 2 implies the second part of Theorem 2 (Hermite's bound).

- Weak LLL Reduction
  - Note that we are only able to prove that $\|\bar{b}_1\|$ reduces in each iteration, but there is no control over the amount by which the norm reduces, and thus it is unclear whether Algorithm 2 is poly-time. To resolve this, we introduce a "slack" ε in Algorithm 2, called the Lovàsz condition. That is, the algorithm only spends time on doing a Lagrange reduction whenever $\|\bar{b}_i\|$ is *significantly* larger than what it could be. Introducing this slack allows us to show that in each iteration the norm must reduce by a factor dependent on ε and argue polynomial number of iterations.
  - **Definition 4.** Let $\bar{B}$ be a lattice basis. For $i \in [n-1]$ and a parameter $\varepsilon > 0$, $(\bar{b}_i, \bar{b}_{i+1})$ satisfies ε-Lovàsz condition if

$$\|\bar{b}_i^*\| \leq (\gamma_2 + \varepsilon) \cdot \|\bar{b}_{i+1}^*\|.$$

  - **Algorithm 3**:
    - *Input*: Basis $(\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$ and a parameter $\epsilon > 0$
    - *Output*: A basis $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq (\gamma_2 + \epsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$
    - Algorithm:
      1. While $\exists i \in [n]$ such that $(\bar{b}_i, \bar{b}_{i+1})$ is not ε-Lovàsz reduced, do:
         1. Find unimodular $\bar{U} \in \mathbb{Z}^{2\times 2}$ such that $(\bar{b}_i', \bar{b}_{i+1}') := \bar{B}_{i:i+1}\bar{U} = (\pi_i(\bar{b}_i), \pi_i(\bar{b}_{i+1}))\bar{U}$ is Lagrange reduced

    2. Update $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_{i-1}, \bar{b}'_i, \bar{b}'_{i+1}, \bar{b}_{i+2}, \cdots, \bar{b}_n)$
  2. Output $\bar{B}$.

- **Theorem 3.** Algorithm 3 *terminates* after $poly(n, \log(\|\bar{B}\|_\infty, 1/\varepsilon))$ iterations and *outputs* $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq (\gamma_2 + \varepsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$ (i.e., an $2^{O(n)}$-approximate solution to SVP).
- *Proof.* Correctness and termination can be argued as for Algorithm 2. Below we focus on bounding the number of iterations. To this end, we use a potential argument. Define

$$P(\bar{B}) := \Pi_{i=1}^n \det(\Lambda_{1:i}) = \Pi_{i=1}^n \|\bar{b}_i^*\|^{n-i+1}$$

  to be potential of a basis $\bar{B}$. Below we will show that $P(\bar{B})$ has a lower bound and that $P(\bar{B})$ decreases by a constant factor in each iteration of Algorithm 3.
  1. To see why the latter is true, observe that $\Lambda_{1:i} \subseteq \mathbb{Z}^n$ and thus $\det(\Lambda_{1:i})$ is an integer and positive thanks to the expression above
  2. First observe that an application of Lagrange reduction on $\bar{B}_{i:i+1}$ only changes $\det(\Lambda_{1:i})$ and leaves the rest of the determinants in the product unchanged.
     - **Homework 3.** Prove the above observation.
  3. Before this application of Lagrange reduction, we have

$$\|\bar{b}_i^*\|^2 > (\gamma_2 + \varepsilon) \cdot \det(\Lambda_{i:i+1}),$$

     and after it we have

$$\|\bar{b}_i^*\|^2 \leq \gamma_2 \cdot \det(\Lambda_{i:i+1}).$$

     Thus the potential reduces by a factor $\rho := \sqrt{\gamma_1/(\gamma_2 + \varepsilon)}$ during this iteration. Note that the initial potential is bounded by

$$\Pi_{i=1}^n \|\bar{b}_i\| = \|\bar{B}\|_\infty^{n(n+1)/2},$$

     where

$$\|\bar{B}\|_\infty := \max_{1 \leq i \leq n} \|\bar{b}_i\|.$$

     Thus the number of iterations N is such that $\|\bar{B}\|_\infty^{n(n+1)/2} \cdot \rho^N \geq 1$, and thus

$$(n(n+1)/2) \log(\|\bar{B}\|_\infty) + N \log(\rho) \geq 0.$$

     Thus $N \leq (n(n+1)) \log(\|\bar{B}\|_\infty) / -2\log(\rho)$. The claim follows by observing that

$$-\log(\rho) = \log(1 + \varepsilon/\gamma_2)/2 = \varepsilon/2\gamma_2 + O(\varepsilon^2).$$

Further Reading

- You can read the recursive variant of Hermite's algorithm in [3], Chapter 2: see Algorithm 5.

2026s-CS758-Lecture07

Topics

- LLL algorithm
- Size-reduction

Sources

1. Intro to Lattice Algorithms and Cryptography, Lecture 4
2. Intro to Lattice Algorithms and Cryptography, Lecture 2
3. Lattices Algorithms and Applications, Lecture 04
4. Lattices Mini Course, Lecture 05

Plan

- Recall/Plan
  - In the last lecture:
    - We saw the *weak* Lenstra-Lenstra-Lovàsz (WLLL) reduction, which solves SVP up to an *exponential* approximation factor. The algorithm is essentially a relaxation of Hermite's reduction in which the Lagrange reduction is done only whenever $\|\bar{b}_i\|$ is *significantly larger* than what it could be, where "significantly larger" is determined by a parameter ε. Recall that Hermite's algorithm itself carries out Lagrange reduction "locally".
    - **Algorithm 1 (WLLL reduction).**
      - *Input*: Basis $(\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$ and a parameter $\varepsilon > 0$
      - *Output*: A basis $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq (\gamma_2 + \varepsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$
      - Algorithm:
        1. While $\exists i \in [n]$ such that $\|\bar{b}_i^*\| > (\gamma_2 + \varepsilon) \cdot \|\bar{b}_{i+1}^*\|$, do:
           1. Find unimodular $\bar{U}_L \in \mathbb{Z}^{2\times 2}$ such that $(\bar{b}_i', \bar{b}_{i+1}') := \bar{B}_{i:i+1}\bar{U}_L$ is Lagrange reduced
           2. Update $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_{i-1}, \bar{b}_i', \bar{b}_{i+1}', \bar{b}_{i+2}, \cdots, \bar{b}_n)$
        2. Output $\bar{B}$.
    - We then argued that WLLL reduction runs for at most a polynomial number of *iterations*. However a polynomial run-*time* is not necessarily guaranteed since an iteration itself might run for super-polynomial time. This happens if:
      1. Any entry in $\bar{B}^*$, which is composed of rational numbers, has large numerator or denominator; or
      2. Any entry in $\bar{B}$, which is composed of integers, is large.
  - In today's lecture:

    - We will resolve the above issues today. We will see that the first is already not an issue in WLLL. Then we will see the (full) LLL reduction, which additionally carries out a "size-reduction" step in order to keep the size of entries of $\bar{B}$ in check.
    - We'll describe LLL reduction using the size-reduction algorithm as a black-box. Then in the next lecture we'll see how to size-reduce using Babai's nearest hyperplane algorithm.
- LLL Reduction
    - Recall that the second condition for an Lagrange-reduced basis (Lecture 05, Definition 1) bounds the Gram-Schmidt coefficient, i.e.,

      $$|\mu_{1,2}| \leq 1/2,$$

      where

      $$\mu_{1,2} := \langle \bar{b}_1^*, \bar{b}_2 \rangle / \|\bar{b}_i^*\|^2.$$

      The notion of size-reduction is essentially a *generalisation* of this condition to higher dimensions (we will see a geometric interpretation of this later). In Lemma 1, we will see why this notion is useful: it guarantees that the norm of each basis vector and its Gram-Schmidt counterpart is "not too large" and this, in turn, implies that their numerator is also "not too large" (show this). In fact everything can be represented using $\mathrm{poly}(n)$ bits.
    - **Definition 1 ([1], Definition 12).** A basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice Λ is said to be *size-reduced* if for all $1 \leq i < j \leq n$,

      $$|\mu_{i,j}| \leq 1/2,$$

      where $\mu_{i,j} := \langle \bar{b}_i^*, \bar{b}_j \rangle / \|\bar{b}_i^*\|^2$.
    - **Lemma 1 ([1], Lemma 14).** Recall the definition of potential function of a lattice $\Lambda(\bar{B}) \subset \mathbb{Z}^n$ from Lecture 06:

      $$\Phi(\bar{B}) := \Pi_{i=1}^n \det(\Lambda_{1:i} = \Lambda(\bar{B}_{1:i})) = \Pi_{i=1}^n \|\bar{b}_i^*\|^{n-i+1.} \leq \det(\Lambda)^n$$

      If $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ is size-reduced, then for all $i \in [n]$,

      $$\log(\|\bar{b}_i\|), \log(\|\bar{b}_i^*\|) \leq \mathrm{poly}(n) \cdot \log(\Phi(\bar{B})) \leq \mathrm{poly}(n) \cdot \log(\det(\Lambda)).$$

    - *Proof.* We first prove the second bound. Since $\bar{B} \subset \mathbb{Z}^n$, for all $1 \leq i \leq n$, $\det(\Lambda_{1:i} = \Lambda(\bar{B}_{1:i})) \geq 1$ (as well an integer) and thus $\det(\Lambda_{1:i}) \leq \Phi(\bar{B})$. Since

      $$\|\bar{b}_i^*\| = \det(\Lambda_{1:i}) / \det(\Lambda_{1:i-1}) \leq \det(\Lambda_{1:i}) \leq \Phi(\bar{B}),$$

      taking log on both proves the second bound. Next, since $\bar{B}$ is size-reduced, we can write any basis vector $\bar{b}_i$ as

      $$\bar{b}_i := \bar{b}_i^* + \sum_{j=1}^{i-1} x_i \bar{b}_j^*$$

for some $-1/2 \leq x_i < 1/2$. It follows that

$$\|\bar{b}_i\| \leq \|\bar{b}_i^*\|^2 + \sum_{j=1}^{i-1} |x_i| \|\bar{b}_j^*\|^2 \leq n \cdot \Phi(\bar{B}).$$

Taking log on both sides proves the first bound.

- LLL reduction keeps the entries of the basis vectors and its Gram-Schmidt counterparts in check by additionally (to WLLL reduction) carrying out size-reduction.
  - **Definition 2 ([1], Definition 13).** A basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ of lattice Λ is said to be *ε-LLL-reduced* if it is
    1. Size reduced
    2. ε-WLLL reduced, i.e., for every $1 \leq i < n$, $(\bar{b}_i, \bar{b}_{i+1})$ satisfies the ε-Lovàsz condition, i.e.,

$$\|\bar{b}_i^*\| \leq (\gamma_2 + \varepsilon) \cdot \|\bar{b}_{i+1}^*\|.$$

- Assuming that a basis can be efficiently size-reduced, we now describe the LLL reduction (with difference from WLLL reduction highlighted). In the theorem that follows, we analyse it.
- **Algorithm 2 (LLL reduction).**
  - *Input*: Basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$ and a parameter $\varepsilon > 0$
  - *Output*: A basis $(\bar{b}_1, \cdots, \bar{b}_n)$ that is ε-LLL-reduced and thus with $\|\bar{b}_1\|^2 \leq (\gamma_2 + \varepsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$
  - Algorithm:
    0. Size-reduce $\bar{B}$
    1. While $\exists i \in [n]$ such that $\|\bar{b}_i^*\| > (\gamma_2 + \varepsilon) \cdot \|\bar{b}_{i+1}^*\|$., do:
       1. Find unimodular $\bar{U}_L \in \mathbb{Z}^{2\times 2}$ such that $(\bar{b}_i', \bar{b}_{i+1}') := \bar{B}_{i:i+1}\bar{U}_L$ is Lagrange reduced
       2. Update $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_{i-1}, \bar{b}_i', \bar{b}_{i+1}', \bar{b}_{i+2}, \cdots, \bar{b}_n)$
       3. Size-reduce $\bar{B}$
    2. Output $\bar{B}$.
- **Theorem 2.** Algorithm 2 *outputs* $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq (\gamma_2 + \varepsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$ in time $T_{LLL} := \mathrm{poly}(n, \log(\|\bar{B}\|_\infty, 1/\varepsilon)$.
- *Proof.* Correctness and termination in $\mathrm{poly}(n, \log(\|\bar{B}\|_\infty, 1/\varepsilon)$ iterations can be argued as in WLLL once you show that size-reduction doesn't affect the potential as we defined in Lecture 06.
  - **Homework 2.** Show that size-reduction does not affect the potential

$$P(\bar{B}) := \Pi_{i=1}^n \det(\Lambda_{1:i}) = \Pi_{i=1}^n \|\bar{b}_i^*\|^{n-i+1},$$

    by showing that the Gram-Schmidt basis is unchanged.
- What remains to show is that the entries of $\bar{B}$ and $\bar{B}^*$ are bounded at all times, and thus each iteration runs in poly-time. To this end, we need to bound the numerator

and denominator of their entries. In the first item below, we bound the denominators of Gram-Schmidt basis $\bar{B}^*$ in terms of the determinant of $\bar{B}$. Note that we have already bounded the norm of size-reduced basis and its Gram-Schmidt counterpart, and thus we are indirectly bounding the numerator. What remains is to show that Lagrange reducing a basis does not blow up the entries: this we prove in the second item below.

1. We first bound the denominators of the Gram-Schmidt orthogonal basis $\bar{B}^*$. Note that this is a general claim, and applies also to WLLL reduction.
   - **Claim 1 ([1], Lemma 11).** If $\bar{B} \in \mathbb{Z}^{n \times n}$ then
   $$\bar{B}^* \in \frac{1}{\det(\bar{B}^t \bar{B})} \mathbb{Z}^{n \times n}.$$
   - *Proof.* We proceed by induction over the dimension n. Consider the projected basis
   $$\bar{B}_{2:n} = (\pi_2(\bar{b}_2), \cdots, \pi_2(\bar{b}_n)).$$
   First observe that the scaled basis
   $$\|\bar{b}_1\|^2 \cdot \bar{B}_{2:n} \in \mathbb{Z}^{n \times (n-1)}$$
   since, by definition, for any $i \in [2, n]$
   $$\pi_2(\bar{b}_i) := \bar{b}_i - \langle \bar{b}_i, \bar{b}_1 \rangle / \|\bar{b}_1\|^2.$$
   On applying the induction hypothesis to $\|\bar{b}_1\|^2 \cdot \bar{B}_{2:n}$, we get
   $$(\|\bar{b}_1\| \cdot \bar{B}_{2:n})^* = \|\bar{b}_1\| \cdot \bar{B}^*_{2:n} \in \frac{1}{\|\bar{b}_1\|^2 \cdot \det(\bar{B}^t_{2:n} \bar{B}_{2:n})} \mathbb{Z}^{n \times (n-1)}.$$
   Since $\bar{B}_{2:n}$ is full rank,
   $$\det(\bar{B}_{2:n} \bar{B}^t_{2:n}) = \Pi_{i=2}^n \|\bar{b}^*_i\|^2.$$
   Since $\bar{b}^*_1 = \bar{b}_1$, we have $\det(\bar{B}\bar{B}^t) = \|\bar{b}_1\|^2 \cdot \det(\bar{B}_{2:n} \bar{B}^t_{2:n})$ and thus
   $$\bar{B}^*_{2:n} \in \frac{1}{\det(\bar{B}\bar{B}^t)} \mathbb{Z}^{n \times n}.$$
   Since each entry $b_{1,j}$ in $\bar{b}_1$ can be expressed as $(b_{1,j} \det(\bar{B}\bar{B}^t)) / \det(\bar{B}\bar{B}^t)$, the claim follows. (Alternatively, $\bar{b}_1$ is orthogonal to each basis vector in $\bar{B}_{2:n}$ and thus $\bar{b}_1 \in 1/\det(\bar{B}\bar{B}^t)\mathbb{Z}^{n \times n}$.)
2. We now prove, assuming Lemma 1, that Lagrange reduction does not blow up the norm of the basis vectors, and hence the size of its entries.
   - **Claim 2 ([1], Lemma 16).** Let $(\bar{b}_1, \bar{b}_2)$ be two linearly-independent vectors, and let $(\bar{b}'_1, \bar{b}'_2) := (\bar{b}_1, \bar{b}_2) \cdot \bar{U}_L$ denote the corresponding Lagrange-reduced basis. Then
   $$\log(\|\bar{b}'_1\|), \log(\|\bar{b}'_2\|) \leq \mathrm{poly}(\log(\|\bar{b}_1\|), \log(\|\bar{b}_2\|)).$$

Moreover, during the Lagrange reduction, all the intermediate basis vectors will also satisfy the above bound.

- *Proof.* We prove the upper bound for $\log(\|\bar{b}_1'\|)$: the proof for $\log(\|\bar{b}_2'\|)$ is similar. Recall from Lecture 05, Theorem 2 that Lagrange reduction on a basis $(\bar{b}_1, \bar{b}_2)$ runs for at most $N_L := O(\log(\|\bar{b}_1\|/\sqrt{\det(\Lambda)})$ iterations. In its ith iteration, the current basis $(\bar{b}_1, \bar{b}_2)$ is multiplied with a unimodular matrix of the form

$$\bar{U}_i := \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \mu_{1,2} \rceil \end{bmatrix},$$

where $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \|\bar{b}_1\|^2$ is the Gram-Schmidt coefficient. By Cauchy-Schwartz inequality, $|\langle \bar{b}_1, \bar{b}_2 \rangle| \leq \|\bar{b}_1\| \|\bar{b}_2\|$ and thus

$$|\lfloor \mu_{1,2} \rceil| \leq \|\bar{b}_2\| / \|\bar{b}_1\|.$$

Moreover, the ℓ_2-matrix norm of $\bar{U}_i$, which is defined as

$$\|\bar{U}_i\| := \max_{\bar{x} \neq 0} \frac{\|\bar{U}_i \bar{x}\|}{\|\bar{x}\|}$$

can be shown to be bounded by

$$\sqrt{\lfloor \mu_{1,2} \rceil^2 + 2} \leq |\lfloor \mu_{1,2} \rceil| + 2.$$

It follows that

$$\|\bar{b}_1'\| \leq \|\bar{b}_1 \bar{U}_L\| = \|\bar{b}_1 \bar{U}_1 \cdots \bar{U}_{N_L}\| \leq \|\bar{b}_1\| \|\bar{U}_1\| \cdots \|\bar{U}_{N_L}\|,$$

which after taking logarithms on both sides, yields

$$\log(\|\bar{b}_1'\|) \leq \log(\|\bar{b}_1\|) + \sum_{i=1}^{N_L} \log(\|\bar{U}_i\|) \leq \log(\|\bar{b}_1\|) + N_L \log(\|\bar{b}_2\| / \|\bar{b}_1\| + 2),$$

which proves the claim.

- ==Size-Reduction ==
  - What remain to answer is how to size-reduce a basis. To this end, we are going to use Babai's Nearest Plane algorithm, which is a close variant of Gram-Schmidt algorithm. We discuss this in Lecture 08. For explaining the BNP algorithm it is instructive to have a geometric interpretation of size-reduction, via Babai's Fundamental Region.
  - Babai's Fundamental Region
    - Recall fundamental parallelepiped of a basis $\bar{B}$ from Lecture 03 (e.g., see $\Lambda((3,1)^t, (1,2)^t)$ in Figure 1).
    - **Definition 2.** Given a basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_m) \in \mathbb{R}^{n \times m}$, the *fundamental parallelepiped* associated with $\bar{B}$ is the region

$$P(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i \bar{b}_i : 0 \leq x_i < 1 \right\}$$

- Recall that the (linear) span of $\Lambda(\bar{B})$ can be "tiled" using $P(\bar{B})$: i.e., it is both a covering and a packing of the span (see formal definition below). In particular, when $m = n$, which we will focus on for the rest of the lecture, the ambient space $\mathbb{R}^n$ can be tiled using $P(\bar{B})$. As a result, as you had to prove in Assignment 1, for each point $\bar{z} \in \mathbb{R}^n$, there exists a *unique* "representative" lattice point $\bar{y}$ such that

$$\bar{z} \in \bar{y} + P(\bar{B}).$$

  - **Example 1.** For $\Lambda((3,1)^t, (1,2)^t)$ from Figure 1, the representative of $(5,5)^t$ is the lattice point $((3,1)^t + 2(1,2)^t = (5,5)^t$.
- More generally, one can talk about fundamental regions, defined below.
  - **Definition 3 ([3]).** Let $\Lambda \subset \mathbb{R}^n$ be a full-rank lattice and let $\mathcal{S}$ be a (measurable) set. We define $\mathcal{S}$ to be
    1. Λ-*packing* if $\forall \bar{y}_1 \neq \bar{y}_2 \in \Lambda$, $(\bar{y}_1 + \mathcal{S}) \cap (\bar{y}_2 + \mathcal{S})) = \emptyset$.
    2. Λ-*covering* if $\Lambda + \mathcal{S} = \mathbb{R}^n$.
    3. Λ-*tiling* or a *fundamental region* of Λ if $\mathcal{S}$ is both Λ-packing and Λ-covering.
- A fundamental region related to $P(\bar{B})$ is the *centred half-open parallelepiped* $P'(\bar{B})$, obtained by "shifting" $P(\bar{B})$ (see Figure 1):

$$P'(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i \bar{b}_i : -1/2 \leq x_i < 1/2 \right\}.$$

  The size-reduction requirement can be seen through a fundamental region called the *orthogonalised centred parallelepiped* $P^*(\bar{B})$, a/k/a *Babai's fundamental region*, which is defined in terms of the Gram-Schmidt basis (see Figure 2 for Babai's fundamental region for $\Lambda((3,1)^t, (1,2)^t)$:

$$P^*(\bar{B}) := P'(\bar{B}^*) = \left\{ \sum_{i=1}^{m} x_i \bar{b}_i^* : -1/2 \leq x_i < 1/2 \right\}.$$

- The size-reduction condition can then be geometrically interpreted as requiring the *orthogonal projections* of $\bar{b}_i$ on to $\bar{b}_1^*, \cdots, \bar{b}_{i-1}^*$ (viz, $\bar{b}_i - \bar{b}_1^*, \cdots, \bar{b}_i - \bar{b}_{i-1}^*$) *to lie in* $P^*(\bar{B})$. As noted above this is a generalisation of the second wristwatch condition.
  - **Example 2.** The lattice $\Lambda((3,1)^t, (1,2)^t)$ from Figures 1 and 2 is already size-reduced since $\mu_{1,2} = 5/10 = 1/2$.

2026s-CS758-Lecture08

Topics

- Size-reduction continued: Babai's Nearest Plane (BNP) Algorithm
- Closest Vector Problem (CVP) and its Friends
- Reducing SVP to CVP

Sources

1. Lattices Algorithms and Applications, Lecture 04
2. Lattices Mini Course, Lecture 05
3. A Gentle Tutorial for Lattice-Based Cryptanalysis

Plan

- Recall/Plan
  - In the last lecture:
    - We introduced *size-reduction* criterion for a basis $\bar{B}$, which *algebraically* requires the corresponding (upper-triangular) Gram-Schmidt matrix $\bar{T}$ to have off-diagonal values less than $1/2$. We then saw how size-reduction prevents the entries in WLLL reduction from blowing up and thus guarantee polynomial run-time.
    - We also discussed what size-reduction means *geometrically* via Babai's fundamental region $P^*(\bar{B})$, which is the fundamental region induced by the Gram-Schmidt basis: a basis $\bar{B} = (\bar{b}_1, \cdots, \bar{b}_n)$ is size-reduced if, for all $i \in [n]$, the orthogonal projection of $\bar{b}_i$ on to the subspace spanned by $(\bar{b}_1, \cdots, \bar{b}_{i-1})$ lies in $P^*(\bar{B})$. Recall that this generalises the second wristwatch condition.
  - In today's lecture:
    - We will see that the geometric interpretation yields a natural size-reduction algorithm, and how it can be computed using Babai's Nearest Plane (BNP) algorithm.
    - In this process, we will introduce the closest vector problem (CVP) and its friends, and see that BNP actually approximately solves CVP.
- Size-reduction continued: Babai's Nearest Plane (BNP) algorithm
  - *Size Reduction.*
    - Recall the *orthogonalised centred parallelepiped* $P^*(\bar{B})$, a/k/a *Babai's fundamental region*, which is defined in terms of the Gram-Schmidt basis:

$$P^*(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i \bar{b}_i^* : -1/2 \leq x_i < 1/2 \right\}.$$

- **Claim 1.** If $\bar{b}_i - \bar{b}_i^* \in P^*(\bar{B})$ for all $1 \le i \le n$, then for all $1 \le j < i \le n$,

$$-1/2 \le \mu_{i,j} < 1/2.$$

- **Homework 1.** Prove Claim 1.
- Given the geometric interpretation of size-reduction in terms of $P^*(\bar{B})$, a natural attempt to find a reduced basis is to replace each basis vector $\bar{b}_i$ with its *reduced counterpart* $\bar{b}_i - \bar{b}_i'$ where $\bar{b}_i'$ is the representative of $\bar{b}_i$'s orthogonal projection $\bar{b}_i - \bar{b}_i^*$ (on to previous subspace) with respect to $P^*(\bar{B})$ (see Figure 1).
- **Algorithm 1 ([1]).**
  - *Input*: Basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$
  - *Output*: A size-reduced basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$
  - Algorithm:
    1. For $i = 2, \cdots, n$ do:
        1. Find the "representative" lattice point of $\bar{b}_i$'s orthogonal projection $\bar{b}_i - \bar{b}_i^*$ with respect to $P^*(\bar{B})$, i.e, $\bar{b}_i' \in \Lambda(\bar{b}_1, \cdots, \bar{b}_{i-1})$ such that

$$\bar{b}_i - \bar{b}_i^* \in \bar{b}_i' + P^*((\bar{b}_1, \cdots, \bar{b}_{i-1}))$$

        2. Update $\bar{b}_i := \bar{b}_i - \bar{b}_i'$
    2. Output $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$
- **Example 1.** Consider $\Lambda((1,2)^t, (3,5)^t)$: see Figure 1. Algorithm 3 proceeds as follows:
  1. First, note that the basis is not size reduced since

$$\mu_{1,2} := \langle (1,2)^t, (3,5)^t \rangle / \|(1,2)\|^2 = 13/5 = 2.6 \not\le 1/2.$$

  2. Since

$$\bar{b}_2^* = (3,5)^t - 2.6(1,2)^t = (0.4, -0.2),$$

     the representative of $\bar{b}_2$'s orthogonal projection

$$\bar{b}_2 - \bar{b}_2^* = (2.6, 5.2) = 3\bar{b}_1 + (-0.4, -0.8)^t$$

     on to (the sub-lattice spanned by) $\bar{b}_1$ is $\bar{b}_1' := 3\bar{b}_1$.
  3. Thus we update $\bar{b}_2 := (3,5)^t - 3(1,2)^t = (0,-1)^t$, and the reduced basis output is $(1,2)^t, (0,-1)^t$.
- **Claim 4.** Algorithm 3 outputs a size-reduced basis.
- *Proof.* First observe that the output is a valid basis since we only carry out elementary column operations in Step 1.2. Next, by definition of $P^*(\bar{B})$, the Gram-Schmidt coefficients of the output basis satisfies the size-reduction condition. Moreover, once we have set a basis vector $\bar{b}_i$ in Step 2.2, it is not disturbed.

- *Babai's algorithm.*
  - What remains is to show how to find the representative lattice point required in Step 1.1. To this send, we use BNP algorithm, stated below. Just like in Lagrange reduction, the idea is to do Gram-Schmidt with rounding. Before describing the algorithm, we explain the geometric intuition behind it (which extends the geometry behind Lagrange reduction).
  - *Geometry behind BNP algorithm (as explained in [2]).*
    - We proceed recursively in the dimensions.
    - A lattice $\Lambda := \Lambda(\bar{b}_1, \cdots, \bar{b}_n)$ can be viewed as a "base" sub-lattice $\Lambda_{1:n-1} := \Lambda(\bar{b}_1, \cdots, \bar{b}_{n-1})$ along with its copies *shifted* by integer multiples of $\bar{b}_n$ (see Figure 2), i.e.,

      $$\{\Lambda_{1:n-1} + x_n \bar{b}_n, x_n \in \mathbb{Z}\}.$$

      In other words, Λ is partitioned into cosets (or affine hyperplanes, hence the name) induced by the base sub-lattice.
    - Now, given a target point $\bar{t} \in \mathbb{R}^n$, what the algorithm tries to do is identify the nearest hyperplane to $\bar{t}$ and it does so based on the Gram-Schmidt algorithm (see Figure 2). To be precise, since the cosets are shifted by $\bar{b}_n$, the algorithm computes the orthogonal projection of $\bar{t}$ on to $\bar{b}_n^*$ (equivalently, the orthogonal complement of $\Lambda_{1:n-1}$), i.e.,

      $$\langle \bar{t}, \bar{b}_n^* \rangle / \langle \bar{b}_n^*, \bar{b}_n^* \rangle,$$

      and to determine the nearest coset it just rounds this value.
    - This process is then recursively carried out over $\Lambda_{1:n-1}$.
  - **Algorithm 2 ([1]).**
    - *Input*: Basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$ and a target point $\bar{t} \in \mathbb{R}^n$
    - *Output*: representative $\bar{B}\bar{x} \in \Lambda$ such that $\bar{t} \in \bar{B}\bar{x} + P^*(\bar{B})$
    - Algorithm:
      1. Set $\bar{e} := \bar{t}$
      2. For $i = n, \cdots, 1$ do:
         1. Compute $x_i := \lfloor \langle \bar{e}, \bar{b}_i^* \rangle / \langle \bar{b}_i^*, \bar{b}_i^* \rangle \rceil$
         2. Set $\bar{e} := \bar{e} - x_i \bar{b}_i$
      3. Output $\bar{t} - \bar{e}$
  - **Example 2 ([3]).** Consider $\Lambda((0,2)^t, (2,1)^t)$ and target point $\bar{t} := (7/2, 7/2)$ (see Figure 3).
    - For $i = 2$, since

      $$\bar{b}_2^* = (2,1)^t - 2/4(0,2)^t = (0,2)^t,$$

      we have

      $$x_2 = \lceil \langle (7/2, 7/2)^t, (2,0)^t \rangle / \langle (2,0)^t, (2,0)^t \rangle \rfloor = \lceil 7/4 \rfloor = 2,$$

and thus the algorithm sets

$$\bar{e} := (7/2, 7/2)^t - 2 \cdot (2,1)^t = (-1/2, 3/2)^t.$$

- For $i = 1$, since $\bar{b}_1^* = \bar{b}_1$, we have

$$x_1 = \lceil \langle (-1/2, 3/2)^t, (0,2)^t \rangle / \langle (0,2)^t, (0,2)^t \rangle \rfloor = \lceil 3/4 \rfloor = 1,$$

and thus the algorithm sets

$$\bar{e} := (-1/2, 3/2)^t - 1 \cdot (0,2) = (-1/2, -1/2).$$

- Thus the algorithm outputs $(7/2, 7/2) - (-1/2, -1/2) = (4,4)$.

- **Claim 2 ([1]).** Algorithm 2 is correct, i.e., it outputs $\bar{y} := \bar{B}\bar{x}$ such that $\bar{t} \in \bar{y} + P^*(\bar{B})$. In particular, at the time of output,

$$\|\bar{e}\|^2 \leq \left(\sum_{i=1}^n \|\bar{b}_i^*\|^2 \right) / 4.$$

- *Proof.* Intuitively, this follows by rounding. Recall that we want to show that $\bar{t} = \bar{B}\bar{x} + \bar{B}^*\bar{e}$ for $\bar{e} = (e_1, \cdots, e_n)$ with $e_i \in [-1/2, 1/2)$. Let $\bar{e}_1, \cdots, \bar{e}_n$ denote the value of $\bar{e}$ set in Step 2 for $i = n, \cdots, i = 1$, respectively. Note that

$$\bar{e}_i = \bar{e} - \sum_{n \geq j > i} \bar{x}_j \bar{b}_j.$$

Let us now consider the component of $\bar{B}^*\bar{e} = \bar{B}\bar{x} - \bar{t}$ along $\bar{b}_i^*$, i.e.,

$$\langle \bar{B}^*\bar{e}, \bar{b}_i^* \rangle = \|\bar{b}_i^*\|^2 \cdot e_i.$$

We also have

$$\begin{aligned}
\langle \bar{B}\bar{x} - \bar{t}, \bar{b}_i^* \rangle &= \langle \sum_{j=1}^n x_j \bar{b}_j - \bar{t}, \bar{b}_i^* \rangle \\
&= \langle x_i \bar{b}_i - (\bar{t} - \sum_{n \geq j > i} x_j \bar{b}_j), \bar{b}_i^* \rangle. \\
&= \langle x_i \bar{b}_i - \bar{e}_i, \bar{b}_i^* \rangle \\
&= x_i \langle \bar{b}_i, \bar{b}_i^* \rangle - \langle \bar{e}_i, \bar{b}_i^* \rangle.
\end{aligned}$$

Now, because we define $x_i := \lfloor \langle \bar{e}_i, \bar{b}_i^* \rangle / \langle \bar{b}_i^*, \bar{b}_i^* \rangle \rceil$, from $\langle \bar{B}^*\bar{e}, \bar{b}_i^* \rangle = \|\bar{b}_i^*\|^2 \cdot e_i.$, we have

$$e_i = x_i \frac{\langle \bar{b}_i, \bar{b}_i^* \rangle}{\|\bar{b}_i^*\|^2} - \frac{\langle \bar{e}_i, \bar{b}_i^* \rangle}{\|\bar{b}_i^*\|^2} = x_i - \frac{\langle \bar{e}_i, \bar{b}_i^* \rangle}{\|\bar{b}_i^*\|^2} \in [-1/2, 1/2],$$

where the second equality follows since by Gram-Schmidt process

$$\langle \bar{b}_i, \bar{b}_i^* \rangle / \|\bar{b}_i^*\|^2 = 1.$$

- Closest Vector Problem and its Friends

- Finally, we will finally see that BNP actually solves a central computational problem on lattices, the closest vector problem (CVP). To define CVP and its friends, we need to define the distance of a point from ambient space to the lattice. Intuitively, it is the radius of the smallest ball centred at $\bar{t}$ that contains a lattice point.
  - **Definition 1.** For a lattice $\Lambda \subset \mathbb{R}^n$ and a target point $\bar{t} \in \mathbb{R}^n$, we define the *distance* of the target point *from the lattice* to be

$$\Delta(\Lambda, \bar{t}) := \min_{\bar{y} \in \Lambda} \|\bar{y} - \bar{t}\|.$$

  - **Example 3.** Consider $\Lambda((0,2)^t, (2,1)^t)$ and target point $\bar{t} := (7/2, 7/2)$ from Example 2 (see Figure 3). The nearest point to $\bar{t}$ turns out to be $(4,4)$, which was actually output by BNP.
- Informally, CVP requires finding the closest lattice point to a given point from the ambient space. In terms of coding theory, you can think of it being the problem of decoding a corrupted codeword to the nearest codeword (of which there may be more than one). Like SVP, we can also define its approximate variants
  - **Definition 2 (CVP: Exact Closest Vector Problem, [1], Definition 2).**
    - *Input.* A basis $\bar{B}$ of lattice Λ and a target point $\bar{t} \in \mathbb{R}^n$
    - *Solution.* A lattice point $\bar{y} \in \Lambda$ such that $\|\bar{t} - \bar{y}\| = \Delta(\Lambda, \bar{t})$.
  - **Definition 3 (CVP$_\alpha$: α-Approximate CVP, [1], Definition 2).**
    - *Input.* A basis $\bar{B}$ of lattice Λ, a target point $\bar{t} \in \mathbb{R}^n$ and an approximation factor $\alpha(\cdot) \geq 1$
    - *Solution.* A lattice point $\bar{y} \in \Lambda$ such that $\|\bar{t} - \bar{y}\| \leq \alpha(n) \cdot \Delta(\Lambda, \bar{t})$.
  - **Pop-Quiz 1.** Do you think CVP reduce to SVP? What about the other direction?
  - **Claim 2 ([3], Theorem 5.3).** SVP$_\alpha$ reduces to exact CVP$_\alpha$.
  - *Proof.* First convince yourself that CVP is equivalent to the following problem, which can be regarded as SVP on cosets (and thus without the constraint to output a non-zero vector):
    - **Definition 4 (CVP$'_\alpha$: α-Approximate CVP variant, [2], Definition 5.1).**
      - *Input.* A basis $\bar{B}$ of lattice Λ, a shift $\bar{t} \in \mathbb{R}^n$ and an approximation factor $\alpha(\cdot) \geq 1$
      - *Solution.* A point $\bar{s} \in \Lambda - \bar{t}$ such that $\|\bar{s}\| \leq \alpha(n) \cdot \Delta(\Lambda, \bar{t})$.
    - **Homework 2.** Prove that for any $\alpha \geq 1$, CVP$_\alpha$ is equivalent to CVP$'_\alpha$.
    - Let's focus on the exact variants as the proof for approximate variants is similar. The proof relies on two key observations:
      1. The shortest vector

$$\bar{s} := \Sigma_{i=1}^n x_i \bar{b}_i$$

      in a lattice $\Lambda(\bar{b}_1 \cdots, \bar{b}_n)$ must have at least one odd coordinate: if, for

contradiction, all its coordinate are even then $\bar{s}/2$ is also in the lattice and is shorter.

2. The set of lattice points whose ith coordinate is odd is a lattice coset: if Λ_i denotes the sub-lattice obtained from Λ by scaling the i th basis vector by a factor of two, i.e.,

$$\Lambda_i := \Lambda(\bar{b}_1, \cdots, \bar{b}_{i-1}, 2\bar{b}_i, \bar{b}_{i+1}, \bar{b}_n),$$

then $\Lambda_i + \bar{b}_i$ is a coset of Λ_i. Thus any Λ can be partitioned into the sub-lattice Λ_i and $\Lambda_i + \bar{b}_i$.

- Together, the two observations imply that there exists an $i \in [n]$ such that $\Lambda_i + \bar{b}_i$ has a lattice point of length at most $\lambda_1(\Lambda)$. Thus the reduction simply invokes the CVP′ oracle on $\Lambda_i + \bar{b}_i$ for each $i \in [n]$, and returns the shortest vector it sees among them.

- It is also interesting to look at the "promise" version of CVP, called bounded-distance decoding (BDD) problem, in which the given target point is promised to be within the "unique decoding radius" $\lambda_1(\Lambda)/2$.
  - **Definition 5 (BDD: Bounded Distance Decoding, [1], Definition 3).**
    - *Input.* A basis $\bar{B}$ of lattice Λ and a target point $\bar{t} \in \mathbb{R}^n$
    - *Promise.* $\Delta(\Lambda, \bar{t}) \leq \lambda_1(\Lambda)/2$.
    - *Solution.* A lattice point $\bar{y} \in \Lambda$ such that $\|\bar{t} - \bar{y}\| \leq \lambda_1/2$.
  - **Pop-Quiz 2.** Does the promise make BDD easier or harder than CVP? It makes the problem easier: think of factoring integers with the promise that one of the factors is less than some bound B.
- What do we know about these problems? (Source)
  - CVP
    - van Emde Boas showed that exact CVP is $\mathbf{NP}$ hard.
    - In general, CVP is considered to be harder than SVP
  - CVP$_\alpha$
    - Like SVP$_\alpha$, CVP$_\alpha$ is also hard to approximate to within constant factors
  - BDD
    - Believed to be easier, and we will base cryptography on it Module II
- *Why BNP solves (approximate) CVP.*
  - First let us consider the case where the target vector is at a distance

$$\Delta(\Lambda, \bar{t}) = \min_{i \in [n]} \|\bar{b}_i\|^2/2$$

from the lattice, which is the case if the basis is "nice": e.g., $\Lambda((0,2)^t, (2,1)^t)$ from Example 2 (Figure 3). It is not hard to see that in this case BNP algorithm solves the exact CVP: the balls around lattice are non-overlapping and entirely contained in Babai's fundamental domain.

- In general, the basis need not be "nice" and BNP algorithm solves CVP$_\alpha$ for

$$\alpha \leq \left(\sum_{i=1}^{n} \|\bar{b}_i^*\|^2 \right)^{1/2} / \min_{1 \leq i \leq n} \|\bar{b}_i^*\|.$$

- This value can be arbitrarily large for example for the lattice below.
  - **Example 5.** Consider $\Lambda((1,3),(2,5))$: see Figure 4.
- But with a more careful analysis, one can prove the following theorem. LLL-reduced basis are examples of basis for which the quantity considered in the theorem is low.
- **Theorem 3 ([3], Theorem 5.6)**: BNP solves CVP$_\alpha$ for a lattice $\Lambda(\bar{b}_1, \cdots, \bar{b}_n)$ with

$$\alpha(n) \leq \sqrt{n+1} \cdot \max_{1 \leq i < j \leq n} \|\bar{b}_j^*\| / \|\bar{b}_i^*\|.$$

- *Proof.* See [2].
  Further Reading

- ARXIV-CSCHA25 STUB

2026s-CS758-Lecture09

Topics

- Application of LLL: Coppersmith's attack on low-exponent RSA

Sources

1. Small solutions to polynomial equations using lattices and applications in cryptography
2. Mathematics of Public Key Cryptography, Chapter 19

Plan

- Recall/Plan Today
  - In previous lectures:
    - We encountered lattice-based computational problems like SVP and CVP, and seen algorithms that approximately solve them in the form of LLL reduction and BNP algorithm.
  - In today's lecture:
    - We will see a concrete application of LLL reduction: cryptanalysis of low-exponent RSA. The general theme is to encode the given problem as a lattice problem in such a way that the approximate shortest vector yields a solution to it.
- RSA Encryption: a Primer
  - Let's refresh public-key encryption (PKE)
    - *Setting:* Comparison with secret-key encryption (SKE):
      - In SKE, Alice and Bob have already met before and exchanged a (symmetric) key k, and that can used to privately send messages to each other
      - In PKE, Alice publishes her public *encryption* key e (on, say, some bulletin board) and Bob uses e to privately send messages to Alice, who can then decrypt using the decryption key d related to e
    - *Syntax:* a triple of algorithms $(\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec})$
      - $(e, d) \leftarrow \mathsf{KGen}(1^\lambda)$: the randomised *key-generation algorithm* KGen takes as input a security parameter λ in unary, and outputs a public encryption key e and a private decryption key d.
      - $C \leftarrow \mathsf{Enc}_e(m)$: the (potentially-randomised) *encryption algorithm* Enc takes as input a message m and outputs a ciphertext c under e
      - $m := \mathsf{Dec}_d(C)$: the deterministic *decryption algorithm* Dec takes as inputs a ciphertext C and a decryption key d, and outputs a message M
    - *Requirements:*

- *Correctness of decryption:* for every λ and m,

$$\Pr_{(e,d)\leftarrow \mathsf{KGen}(1^\lambda)}[\mathsf{Dec}_d(\mathsf{Enc}_e(m)) = 1.$$

- *One-way security:* No efficient (i.e., probabilistic polynomial-time) adversary A can recover a message(of sufficient entropy) from a ciphertext, i.e., for every efficient A and (random) m,

$$\epsilon(\lambda) := \Pr_{\substack{(e,d)\leftarrow \mathsf{KGen}(1^\lambda)\\ c\leftarrow \mathsf{Enc}_e(m)\\ m'\leftarrow \mathsf{A}(e,c)}}[m' = m]$$

  is negligible.

- In this lecture, we will look at restricted RSA encryption, called low-exponent RSA, where the public encryption $e = 3$:
  - Scheme
    - $\mathsf{KGen}(1^\lambda)$:
      1. Sample two distinct primes P and Q of length $\approx \lambda^3$ and compute the semiprime $N := PQ$.
      2. Output $(e = 3, N)$ as the encryption key and $(d := 3^{-1} \bmod (P-1)(Q-1), N)$ as the decryption key
    - $\mathsf{Enc}_N(m)$: output $c := m^3 \bmod N$
    - $\mathsf{Dec}_d(c)$: output $c^d \bmod N$
  - Correctness of decryption can be argued using basic number theory, and security can be based on RSA assumption.
- Low-exponent RSA with known padding
  - Sometimes RSA encryption is carried out on messages of the form $m := p + r$, where p is publicly known (e.g., a known prefix such as the string "The PIN is ") and r is small (e.g., a six-digit PIN). In such cases, RSA encryption algorithm is captured by the function

$$f_{p,N}(x) := x^3 + 3px^2 + 3px + p^3 \bmod N$$

    which is a monic univariate polynomial of degree three, generally represented as

$$f(x) = x^3 + a_2x^2 + a_1x + a_0 \bmod N.$$

  - Thus, for breaking security of this variant of RSA, we need to find a small root r of $f(x)$ modulo N. A few comments about this task are in order:
    - Finding roots of polynomials modulo a prime p (i.e., the field $(\mathbb{Z}_q, +, \cdot)$) is possible in polynomial time: see Berlekamp's algorithm and Cantor-Zassenhaus algorithm. Similarly, finding (integer) roots of polynomials over reals is possible: see Newton's method. In fact, looking ahead, we

are going to reduce to the task of finding roots of polynomials over reals.
    - Thus the hardness of computing roots of a polynomial only kicks in due to the modulus being a semiprime and its factors being unknown (which are again hard to compute given only N): given the factors P and Q, it again is easy to find the roots via Chinese remaindering theorem.
    - Without the restriction on r being small, the problem is believed to be computationally infeasible (this is related to the RSA problem).
- Coppersmith's Attack
  - It must by now be clear that finding small roots of polynomials modulo N is well-motivated. In this context, Coppersmith proved the following:
  - **Theorem 1 (Theorem 3.2.1 in [1]).** Given a monic univariate polynomial $f(x) \in \mathbb{Z}(x)$ of degree d represented as

    $$f(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$$

    and a modulus $N \in \mathbb{Z}$, there is an algorithm to find all small roots, i.e., $r \in \mathbb{Z}$ satisfying

    $$f(r) = 0 \bmod N$$

    and

    $$|r| \leq N^{1/d}.$$

    Moreover, the algorithm runs in time $\mathrm{poly}(\log(N), d)$.
  - To prove Theorem 1, we take the following high-level approach:
    1. We encode the given polynomial $f(x)$ using an appropriately constructed lattice Λ which, looking ahead, will be of dimension $n := d + 1$. The key property that Λ will satisfy is that its points will encode polynomials and each such polynomial will also have r as a root.
    2. We will use LLL to find a short vector $(a'_d, \cdots, a'_0) \in \Lambda$, encoding a polynomial

       $$f'(x) := a'_d x^d + a'_{d-1}x^{d-1} + \cdots + a'_1 x + a'_0.$$

       Recall that (for appropriately set ε) LLL outputs a basis in which the first vector has ℓ_2 norm at most

       $$2^{(n-1)/4} \cdot \det(\Lambda)^{1/n}.$$

       We will construct Λ in such a way that

       $$|f'(r)| < N \qquad (1)$$

       holds (given $|r| \leq N^{1/d}$), which ensures that $f'(r) = 0$ holds *also over integers*! From our earlier discussion, it is then possible to extract r.
       - **Example 1 ([2], Example 19.1.1).** Consider $f(x) := x^2 + 33x + 21$ and $N := 17 \cdot 19 = 323$. Note that 3 is a solution modulo 323, but not over

integers. However, for

$$f'(x) := 9f(x) - N(x-6) = 9x^2 - 26x - 3$$

3 is a solution also over integers, and this can be found (say) using Newton's method or simply using the quadratic formula.

3. When exactly does the condition in Eq.1 hold? We need the following value to be at most N

$$\begin{aligned} |f'(r)| &= |a'_d r^d + a'_{d-1} r^{d-1} + \cdots + a'_1 r + a'_0| \\ &\leq |a'_d r^d| + |a'_{d-1} r^{d-1}| + \cdots + |a'_1 r| + |a'_0|, \\ &= \|(a'_d r^d, a_{d-1} r^{d-1}, \cdots, a'_0\|_1 \qquad (2) \end{aligned}$$

where if you recall (from Quiz 1) $\|\cdot\|_1$ denotes the ℓ_1-norm. As a result (1) holds, e.g., when

$$|a'_i r^i| < N/(d+1), \quad i \in [0, d].$$

- Let's see in details how the above steps work. A natural way to encode a polynomial as a vector is via its coefficients, i.e.,

$$f(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

can be encoded by

$$\bar{b}_{f(x)} := (1, a_{d-1}, \cdots, a_1, a_0)^t.$$

Since we want all the lattice points all to have r as a root, we need to consider transformations to $f(x)$ which *preserve roots* modulo N.

- **Pop Quiz 1.** Can you think of such transformations?
  1. Adding a monomial $g_i(x) := Nx^i$, since $g(r) = 0 \bmod N$ implies $f(r) + g_i(r) = 0 \bmod N$.
  2. Multiplying a monomial x^i, since $r^i \cdot f(r) = r^j \cdot 0 = 0 \bmod N$.
  3. Exponentiating, since $(f(r))^i = 0^i = 0 \bmod N$.
- As a first attempt let's consider the lattice generated by $f(x)$ and $g_i(x)$s, which for the case of $d = 3$ is generated by the basis

$$\bar{B} := \begin{bmatrix} 1 & & & \\ a_2 & N & & \\ a_1 & & N & \\ a_0 & & & N \end{bmatrix}.$$

Note that the determinant of this (triangular) matrix is N^3.

- **Pop Quiz 2.** What happens we run LLL on $\Lambda := \Lambda(\bar{B})$ to obtain a basis with first basis vector

$$(\bar{b}_{f'(x)} = (a'_3, \cdots, a'_0))$$

encoding $f'(x) = a'_3x^3 + a'_2x^2 + a'_1x + a'_0$? In particular, although we are guaranteed that $f'(r) = 0 \bmod N$, when does $f'(r) = 0$ hold also over integers?

1. First, note that LLL bounds the basis vector in terms of ℓ_2 norm, while our requirement in (2) is in terms of the ℓ_1 norm. However this is not an issue since ℓ_2-norm and ℓ_1-norm are weakly related:
   - **Homework 1.** For any $\bar{z} \in \mathbb{R}^n$, show that $\|\bar{z}\|_1 \leq \sqrt{n} \cdot \|\bar{z}\|_2$.
   - Hint. Use Cauchy-Schwartz inequality.
     Thus LLL outputs a basis in which

$$\|\bar{b}_{f'(x)}\|_1 \leq \sqrt{n} \cdot 2^{(n-1)/4} \cdot \det(\Lambda)^{1/n}.$$

2. The *more important* issue is that LLL only guarantees an upper bound on size of the coefficients of $\bar{b}_{f'(x)}$ in terms of $\|(a'_3, a'_2, a'_1, a'_0)\|_1$ and not the *evaluations*, which is what is required in (2).

- *How do you resolve this?* Intuitively, by considering the *scaled* lattice Λ_R generated by $f(Rx)$ and $g_i(Rx)$, where $R \in \mathbb{N}$ is an upper bound on the size of r. Thus we consider the following scaled basis

$$\bar{B}_R := \begin{bmatrix} R^3 & & & \\ a_2R^2 & NR^2 & & \\ a_1R & & NR & \\ a_0 & & & N \end{bmatrix}$$

and the lattice $\Lambda_R := \Lambda(\bar{B}_R)$ and then scale back LLL's output. Let's see why setting $R := N^{1/d}$ as in the theorem statement works. Note that since $\bar{B}$ is triangular, its determinant is

$$\det(\Lambda_R) = N^3R^6$$

and in general for degree d,

$$\det(\Lambda_R) = N^dR^{d(d+1)/2}.$$

Thus for appropriately set ε in LLL for the basis vector $\bar{b}_{f'(Rx)}$,

$$\|\bar{b}_{f'(Rx)}\|_1 \leq \sqrt{n} \cdot 2^{(n-1)/4} \cdot \det(\Lambda_R)^{1/n} = \sqrt{d+1} \cdot 2^{d/4} \cdot (N^dR^{d(d+1)/2})^{1/(d+1)}.$$

For the specific example above, $d = 3$ is a small constant and can be ignored, and we roughly require $(N^3R^6)^{1/4} < N$ and thus

$$R < N^{1/6}.$$

More generally, the above analysis will yield $R < N^{1/2d}$, but with a more careful analysis using a slightly different lattice, the bound can be improved to $R < N^{1/d}$ as in the theorem statement. In particular, you will need to use the other two root-preserving transformations we considered above.

- **Homework 2.** Complete proof of Theorem 1 by showing that $R < N^{1/d}$ suffices.

- **Example 2 ([2], Example 19.1.6).** Let $f(x) := x^3 + 10x^2 + 5000x - 222$ and $N := 10001$. One can check that $r = 4$ is a root modulo M. Since $r < M^{1/3}$, the attack should work when setting $R := 10$. Note that basis matrix is

$$\bar{B} := \begin{bmatrix} R^3 & & & \\ 10R^2 & NR^2 & & \\ 5000R & & NR & \\ -222 & & & N \end{bmatrix}.$$

  Running LLL yields a reduced basis with first vector $(444, 10, -2000, -2000)$ which corresponds to the scaled polynomial

$$f'(Rx) := -2(Rx)^3 - 20(Rx)^2 + (Rx) + 444$$

  and thus the scaled-back polynomial is

$$f'(x) := -2x^3 - 20x^2 + x + 444,$$

  which has $r = 4$ as a short root.

- Other Attacks
  - Coppersmith's attack can be extended to factorise integers of the form $p^a q$ for a "large" a
  - LLL can be used to break subset sum and knapsack problem, which we will recall below.
    - Search version of subset-sum problem (SSP) (over integers):
      - Input: list of non-negative integers $x_1, \cdots x_n \in \mathbb{N}$, and a target $T \in \mathbb{N}$
      - Solution: a subset $I \subseteq [n]$ such that $\sum_{i \in I} x_i = T$.
      - Note that n denotes the list size and *not* the length of the input
    - *Special* case of knapsack problem:
      - Input: list of (positive) weights $w_1, \cdots, w_n \in \mathbb{N}$ and capacity $T \in \mathbb{N}$,
      - Solution: a maximal packing, i.e., a subset $I \subseteq [n]$ such that:
        1. $\sum_{i \in I} w_i \leq T$ and
        2. For every other subset $I' \subseteq [n]$, $\sum_{i \in I'} w_i \leq \sum_{i \in I} w_i$.
    - What LLL solves is "low-density" subset-sum problem where density relates to size of the entries x_i relative to n.

Further Reading

- A Gentle Tutorial for Lattice-Based Cryptanalysis , for further attacks using LLL
- Factoring N=p^rq for Large r, for extension of Coppersmith's attack to factorisation

2026s-CS758-Lecture10

Topics

- Short integer solution (SIS) problem
- One-way function (OWF) and collision-resistant hash function (CRHF) from SIS

Sources

1. The Short Integer Solutions Problem and Cryptographic Applications, by Micciancio
2. Lattices in Computer Science, Lecture 8, by Regev
3. A Decade of Lattice Cryptography, by Peikert

Plan

- Recall/Plan Today
  - *In Module I:*
    - We defined lattices and studied several lattice invariants like determinant and successive minima, and then bounded its values.
    - We also looked at corresponding computational problems like SVP and CVP. We saw that solving them (even approximately) is hard in general. By hard, we were implicitly referring to their *worst-case* hardness, i.e., for every algorithm there exists hard instances of SVP/CVP.
  - *In Module II:*
    - We will look at *average-case* hardness of lattice problems: i.e., we will come up with hard *distributions* over lattice problems, i.e., for every algorithm a random sample from this distribution is hard to solve. In particular, we will introduce the short integer solution (SIS) and learning with errors (LWE) problems. Both these problems can be viewed as *average-case* versions of CVP. The motivation for looking at average-case hardness is cryptography, where we need the ability to sample random instances (e.g., think of key of a hash function).
  - *In today's lecture:*
    - We will introduce SIS and use it to construct (keyed) OWF and CRHF. In this process, we will see that SIS can be regarded as a special case of SVP/CVP.
    - In the next lecture, we will see that the average-case hardness of SIS can be based on worst-case hardness of SIVP (from Lecture 05). Recall that such worst-case to average-case reduction is one of the four selling points of lattice-based cryptography (with the others being post-quantum guarantee, efficient implementations and expressibility).
- Defining SIS
  - *Worst-case vs average-case hard search problem.*

- Search problems can be defined with respect to (binary) relations as follows: the search problem corresponding to a relation $\mathcal{R} \subseteq \{0,1\}^* \times \{0,1\}^*$ consists of instance-solution pairs, where
  - Instance: $x \in \{0,1\}^*$ for which there exists a witness $w \in \{0,1\}^*$ such that $(x, w) \in \mathcal{R}$
  - Solution: $w \in \{0,1\}^*$ such that $(x, w) \in \mathcal{R}$
- **Example 1.** The relation $\mathcal{S}$ corresponding to semiprime factorisation is

$$\mathcal{S} := \{(N, P) : 1 < P < N, P|N\}.$$

- **Definition 1.** A search problem corresponding to a relation $\mathcal{R}$ is hard to solve in the *worst-case* if for every poly-time algorithm A, there exists $(x, w) \in \mathcal{R}$ such that $(x, \mathsf{A}(x)) \notin \mathcal{R}$.
  - Note that this definition is obtained by *negating* the definition of $\mathcal{R}$ being *easy to solve*, viz., there exists a poly-time algorithm A such that for every $(x, w) \in \mathcal{R}$, $(x, \mathsf{A}(x)) \in \mathcal{R}$.
  - **Example 2.** Factoring semiprimes is hard to solve in the worst case if for every poly-time algorithm A, there exists a semiprime $N := PQ$ such that $\mathsf{A}(N)$ does not output P.
- **Definition 2.** A search problem defined by a relation $\mathcal{R}$ is hard to solve *on the average* if there exists a distribution $R := \{R_n\}_{n \in \mathbb{N}}$ over $\mathcal{R}$ such that for every PPT algorithm $\mathcal{A}$, the following is negligible

$$\epsilon(n) := \Pr_{\substack{(x,w) \leftarrow R_n \\ w' \leftarrow \mathsf{A}(x)}} [(x, w') \in \mathcal{R}]$$

  - **Example 3.** Factoring semiprimes is hard to solve on the average if there exists a distribution $S := \{S_n\}_{n \in \mathbb{N}}$ over $\mathcal{S}$ such that for every poly-time algorithm A, the following is negligible

$$\epsilon(n) := \Pr_{\substack{N \leftarrow S_n \\ P \leftarrow \mathsf{A}(N)}} [1 < P < N, P|N].$$

    - Note that the distribution induced by $N := PQ$ for random primes P and Q of size roughly $n/2$ is believed to be a hard distribution.

- *Defining SIS.* Informally, (homogeneous) SIS requires finding a short non-zero vector $\bar{e}$ in the kernel of a compressing random matrix $\bar{A}$. More generally, *inhomogeneous* SIS requires finding a short inverse $\bar{e}$ of a random $\bar{s}$ under $\bar{A}$. From a coding theory point of view, this is exactly the problem of syndrome decoding: i.e., given the syndrome $\bar{s} := \bar{A}\bar{e}$ of a parity-check matrix $\bar{A}$, find the (short) error $\bar{e}$. One way to quantify "shortness" is by requiring $\bar{e}$ to have binary entries:
  - **Definition 3 (Homogeneous SIS, [1]).**
    - *Parameters:* dimensions $n, m \in \mathbb{N}$ and a modulus $q \in \mathbb{N}$. To *guarantee* a solution, we typically set $m > n\lceil \log(q) \rceil$ as then

$$|\{0,1\}^m| = 2^m > 2^{n|\log(q)|} > q^n = |\mathbb{Z}_q^n|,$$

and a solution is guaranteed by the pigeonhole principle.

- *Input:* $\bar{A} \leftarrow \mathbb{Z}_q^{n\times m}$
- *Solution:* $\bar{e} \in \{0,1\}^m \setminus \{0^m\}$ such that $\bar{A}\bar{e} = 0^n \bmod q$

- **Definition 4 (Inhomogeneous SIS, [3]).**
  - *Parameters:* $n, m, q \in \mathbb{N}$ as in Definition 3
  - *Input:* $\bar{A} \leftarrow \mathbb{Z}_q^{n\times m}$, $\bar{u} \leftarrow \mathbb{Z}^n$
  - *Solution:* $\bar{e} \in \{0,1\}^m$ such that $\bar{A}\bar{e} = \bar{s} \bmod q$
- More generally, shortness can be phrased in terms of the ℓ_2 norm:
- **Definition 4 (Homogeneous SIS, generalised, [3]).**
  - *Parameters:* dimensions $n, m \in \mathbb{N}$, a modulus $q \in \mathbb{N}$ and a norm-bound $B \in \mathbb{N}$. To *guarantee* a solution, set $m \geq n\lceil\log(q)\rceil$ and $B \geq \sqrt{n\lceil\log(q)\rceil}$.
  - *Input:* $\bar{A} \leftarrow \mathbb{Z}_q^{n\times m}$
  - *Solution:* $\bar{e} \in \mathbb{Z}^m \setminus \{0^m\}$ such that $\bar{A}\bar{e} = 0^n \bmod q$ and $\|\bar{s}\| \leq B$.
- Note that the problem is trivial for $B \geq q$ since, e.g., $(q, 0, \cdots, 0)^t$ is a solution.

- Connection to Lattices via CVP
  - At first glance, it is unclear what SIS has to do with lattices. In this section, we will see why homogeneous/inhomogeneous SIS can be regarded as SVP/CVP in a dual lattice defined by $\bar{A}$. To this end, we first recall dual lattice, and study dual lattice of a special class of lattices called q-ary lattices.
  - *Dual lattice.*
    - **Definition 5.** The dual of a (primal) lattice Λ is

$$\Lambda^* := \{\bar{z} \in \mathrm{span}(\Lambda) : \langle \bar{z}, \bar{y}\rangle \in \mathbb{Z} \text{ for all } \bar{y} \in \Lambda\}.$$

    - In Assignment 1, you must have already seen several examples of lattices and their dual.
      - **Pop Quiz 1.** What is the dual of following lattices?
        1. Integer lattice $\mathbb{Z}^n$. It is its own dual.
        2. Rotation of $\mathbb{Z}^n$ by 30°. Scaling doesn't affect duality.
        3. Scaled integer lattice $q\mathbb{Z}^n$. It is the integer lattice scaled by $1/q$. See Figure 1 for the dual of $2\mathbb{Z}^n$.
    - The definition is not very intuitive at first glance, but its roots are in the Fourier transform, as we now explain. An alternative way to think of the dual lattice is as the set of all linear functions from Λ to $\mathbb{Z}$. Recall that the set of all linear functions from Λ to $\mathbb{R}$ is

$$\{\phi_{\bar{z}} : \Lambda \to \mathbb{R}\}_{\bar{z}\in\mathrm{span}(\Lambda)},$$

where

$$\phi_{\bar{z}}(\bar{y}) := \langle \bar{z}, \bar{y}\rangle,$$

and each function can thus be represented using a vector $\bar{z} \in \mathrm{span}(\Lambda)$. The dual lattice is a *subset* of these $\bar{z}$s for which the corresponding linear function has outputs over $\mathbb{Z}$. It is not hard to see that such functions form a lattice.

- Given this interpretation, a primal lattice Λ and its dual Λ^* are related via the Fourier Transform: the primal lattice lives in the "evaluation" domain while the dual lattice lives in the "function/frequency" domain. For an analogy, consider representing polynomials using the Lagrange (i.e., evaluation) versus the monomial basis: the primal lattice corresponds to the Lagrange basis, while the dual lattice corresponds to the monomial basis. For for a physical interpretation in terms of diffraction see this math.stackexchange post.

- *q-ary lattice.* Recall that an integer lattice is a lattice generated by an integer basis. Informally, a q-ary lattice is an integer lattice that is *periodic* modulo q: i.e., if $\bar{y} \in \Lambda$ and $\bar{y} = \bar{y}' \bmod q$ then $\bar{y}' \in \Lambda$.
  - **Definition 6 ([3], Definition 1).** A lattice $\Lambda \subset \mathbb{R}^n$ is called an integer lattice if $\Lambda \subseteq \mathbb{Z}^n$. An integer lattice Λ is said to be q-ary if $q\mathbb{Z}^n \subset \Lambda$.
  - **Example 3 (q-ary lattice underlying SIS).**
    - The lattice defined by the integer *(row)span* of a SIS matrix $\bar{A} \in \mathbb{Z}_q^{n \times m}$, i.e.,

$$\Lambda_q(\bar{A}) := \{\bar{x}^t \bar{A} \bmod q : \bar{x} \in \mathbb{Z}^n\}.$$

    - The lattice defined by the kernel of a SIS matrix $\bar{A} \in \mathbb{Z}_q^{n \times m}$, i.e.,

$$\Lambda_q^{\perp}(\bar{A}) := \{\bar{e} \in \mathbb{Z}^m : \bar{A}\bar{e} = 0^n \bmod q\}.$$

    - Note that these two lattices are different for same $\bar{A}$. In terms of coding theory, these two lattices correspond to the code defined with $\bar{A}$ as the generator matrix and parity-check matrix respectively.
  - In Exercises 1 and 2, you will show that the lattices from Example 3 are indeed lattices and that up to scaling, they are dual of each other.
    - **Exercise 1.** Show that $\Lambda_q(\bar{A})$ and $\Lambda_q^{\perp}(\bar{A})$ are lattices.
    - **Exercise 2.** Show that

$$(\Lambda_q(\bar{A}))^* = (1/q)\Lambda_q^{\perp}(\bar{A})$$

      and

$$(\Lambda_q^{\perp}(\bar{A}))^* = (1/q)\Lambda_q(\bar{A}).$$

  - From the definition of (homogeneous) SIS, and the characterisation of q-ary lattice above, it must be clear that SIS corresponds to SVP in $\Lambda_q^{\perp}(\bar{A})$. To understand how inhomogeneous SIS corresponds to CVP in the same lattice, we proceed via OWFs.
- *One-way function from SIS.*

- Recall the definition of (keyed) one-way function (OWF).
  - **Definition 6.** An efficiently-computable function family

    $$f := \{f_k : \mathcal{D}_k \to \mathcal{R}_k\}_{k \subseteq \{0,1\}^*}$$

    with key-sampling algorithm KGen is *one-way* if for all PPT inverters Inv, the following is negligible

    $$\epsilon(\lambda) := \Pr_{\substack{k \leftarrow \mathsf{KGen}(1^\lambda) \\ x \leftarrow \mathcal{D}_k}} [\mathsf{Inv}(f_k(x)) \in f_k^{-1}(x)].$$

  - **Example 4.** The following is a OWF based on discrete-log assumption modulo a prime p:

    $$f_{p,g} : \mathbb{Z}_p \to \mathbb{Z}_p^*,$$

    defined as

    $$f_{p,g}(x) := g^x \bmod p.$$

    The prime p and the generator g form the key, and the function is simply exponentiation modulo p
- The following is a very natural construction of OWF from SIS, called Ajtai's function. As we will see in the next lecture, Ajtai showed that the function is one-way based on worst-case hardness of SIVP.
  - **Construction 1 (OWF from inhomogeneous SIS).**
    - *Key:* SIS matrix $\bar{A} \leftarrow \mathbb{Z}_q^{n \times m}$, with $n, m, q \in \mathbb{N}$ as in Definition 3
    - *Input:* $\bar{e} \in \{0,1\}^m$
    - Output: $f_{\bar{A}}(\bar{e}) := \bar{A}\bar{e} \bmod q$
  - To see the connection between CVP and inhomogeneous SIS, we will pose the task of breaking the OWF from Construction 1 as a lattice problem:
    1. First observe that finding a non-short solution to $\bar{A}\bar{e} = \bar{s}$ is *easy* by Gaussian elimination. Let $\bar{t} \in \mathbb{Z}_q^m$ denote one such solution.
    2. Now note that *all* solutions to $\bar{A}\bar{e} = \bar{s}$ are of the form

       $$\bar{t} + \Lambda_q^\perp(\bar{A}),$$

       i.e., they form a coset of the the dual lattice defined by the kernel. See Figure 2.
    3. Thus it boils down to finding a *short vector* in the coset, which from Lecture 08 is the CVP problem rephrased! Thus equivalently, breaking the OWF in Construction 1 boils down to looking for a lattice vector $\bar{y} \in \Lambda_q^\perp(\bar{A})$ that is close to $\bar{t}$.
  - To summarise, inhomogeneous SIS on $\bar{A}$ can be seen as the average-case version of CVP on the dual lattice $\Lambda_q^\perp(\bar{A})$.

- Cryptography from SIS
  - We have already seen in Construction 1 how inhomogeneous SIS implies a (keyed) OWF via Ajtai's function. Observe that for the parameters in consideration, Ajtai's function is already compressing, and can be shown to be collision resistant under homogeneous SIS assumption.
  - *Compression and hashing.*
    - Let us first recall the definition of CRHF.
      - **Definition 7.** An efficiently-computable function family

        $$H := \{H_\lambda : \mathcal{K}_\lambda \times \{0,1\}^* \to \{0,1\}^\lambda\}_{\lambda \in \mathbb{N}}$$

        with key-sampling algorithm KGen is *collision-resistant* if for all PPT collision finders F, the following is negligible

        $$\epsilon(\lambda) := \Pr_{\substack{k \leftarrow \mathsf{KGen}(1^\lambda) \\ (x_1, x_2) \leftarrow \mathsf{F}(k)}} [H_k(x_1) = H_k(x_2), x_1 \neq x_2].$$

    - Given the generalised definition of (homogeneous) SIS (Definition 5), it is easy to construct a collision-resistant *compression* function, where the domain is $\{0,1\}^{\ell(\lambda)}$ for some $\ell(\lambda) > \lambda$. Given a collision-resistant compression function, one can obtain a CRHF e.g., using Merkle-Damgård construction.
      - **Construction 2 (CRHF from homogeneous SIS).**
        - *Key:* $\bar{A} \leftarrow \mathbb{Z}_q^{n \times m}$ as in Definition 4, with $\lambda = n$
        - *Input:* $\bar{e} \in \{0,1\}^m$
        - Hash function: $H_{\bar{A}}(\bar{e}) := \bar{A}\bar{e} \bmod q$
      - **Theorem 1.** If SIS with parameters as defined in Definition 4 is hard on average, then $H_{\bar{A}}$ is CRHF.
      - *Proof.* Suppose $\bar{e}_1 \neq \bar{e}_2 \in \{0,1\}^m$ be a collision under $H_{\bar{A}}$, i.e.,

        $$\bar{A}\bar{e}_1 = \bar{A}\bar{e}_2 \bmod q.$$

        It follows that $\bar{A}(\bar{e}_1 - \bar{e}_2) = 0 \bmod q$, and moreover it is short since $\bar{e}_1 - \bar{e}_2 \in \{-1, 0, 1\}^m$ and $\|\bar{e}_1 - \bar{e}_2\| \leq \sqrt{m} < B$.
  - *Regularity and commitment.* We will see that Ajtai's function also yields a commitment scheme thanks to its output being close to uniformly random. You will construct the commitment scheme as a problem in Assignment 2.
  - *Linearity and digital signature*. We will see this in a later lecture how it is possible to construct Schnorr-like digital signatures rely on linearity of Ajtai's function.

Further Reading

-

2026s-CS758-Lecture11

Topics

- Worst-case to average-case reduction for SIS

Sources

1. The Short Integer Solutions Problem and Cryptographic Applications, by Micciancio
2. Advanced Topics in Cryptography: Lattices (Fall 2015), Lectures [11](Lecture 11) and [12] (Lecture 11), by Vaikuntanathan
3. Worst-case to Average-case Reductions based on Gaussian Measures, Micciancio and Regev

Plan

- Recall/Plan Today
  - In the last lecture:
    - We introduced SIS. Let's recall the bounded version of the definition:
      - **Definition 1 (Generalised homogeneous SIS, Definition 3 in Lecture 10).**
        - *Parameters:* dimensions $n, m \in \mathbb{N}$, a modulus $q \in \mathbb{N}$ and a norm-bound $B \in \mathbb{N}$, which we set to be $q \approx n^2$, $m \geq n\lceil \log(q) \rceil$ and $B \geq \sqrt{m}$.
        - *Input:* $\bar{A} \leftarrow \mathbb{Z}_q^{n \times m}$
        - *Solution:* $\bar{e} \in \mathbb{Z}^m \setminus \{0^m\}$ such that $\bar{A}\bar{e} = 0^n \bmod q$ and $\|\bar{s}\| \leq B$.
      - The problem can be interpreted in a couple of ways:
        - Find a short non-zero vector $\bar{e}$ in the kernel of a compressing random matrix $\bar{A}$. More generally, find a short inverse $\bar{e}$ of a random $\bar{s}$ under $\bar{A}$.
        - From a coding theory point of view, it is exactly the problem of syndrome decoding: i.e., given the syndrome $\bar{s} := \bar{A}\bar{e}$ of a (short) error $\bar{e}$ under a parity-check matrix $\bar{A}$, find $\bar{e}$.
    - We then constructed (keyed) OWF and CRHF under SIS assumption via *Ajtai's function*:
      - **Construction 1 (OWF from inhomogeneous SIS, Construction 1 in Lecture 10).**
        - *Key:* SIS matrix $\bar{A} \leftarrow \mathbb{Z}_q^{n \times m}$, with parameters as in Definition 1
        - *Input:* $\bar{e} \in \mathbb{Z}^n$, with $\|\bar{e}\| < B$
        - Output: $\bar{s} := \bar{A}\bar{e} \bmod q$
      - We saw that inverting the problem is hard since it is essentially CVP in the dual lattice $\Lambda_q^{\perp}(\bar{A})$.

- *A detour.* The above construction yields a general *template* to construct OWF from *any* lattice, which you can think of as *generalised* Ajtai's function:
  - **Construction 2 (OWF from any integer lattice, [1]).**
    - *Key:* Lattice $\Lambda = \Lambda(\bar{B})$
    - *Input:* $\bar{e} \in \mathbb{Z}^n$, with $\|\bar{e}\| < B$
    - Output: $\bar{s} := \bar{e} \bmod \Lambda = \bar{e} \bmod P(\bar{B})$
  - How can $\bar{e} \bmod \Lambda$ be computed?
    - Recall that if $\bar{e} \in \bar{y} + P(\bar{B})$ for the unique representative $\bar{y} \in \Lambda$, then $\bar{e} \bmod \Lambda := \bar{e} - \bar{y}$ (see Figure 1).
    - Alternatively, $\bar{e} \bmod \Lambda$ can be computed using the basis of the dual lattice $\Lambda^* = \Lambda(\bar{D})$ as

$$\bar{e} \bmod \Lambda = \bar{B} \cdot (\langle \bar{D}, \bar{e} \rangle \bmod 1).$$

      - Recall this is how we computed syndrome in the the definition of SIS: you should think of $\bar{A} = \bar{D}$ and thus the lattice is consideration is the one defined by $\bar{A}$'s kernel.
  - **Homework 1.** Show that the two ways to compute $\bar{s}$ yield the same value.
    - Solution by PS: 2026s-CS758-Lecture 11-HW1 PS
- Looking ahead, viewing one-wayness of lattices through the lens of Construction 2 is very useful. Depending on the value set for bound B, we get either SIS or LWE (see Figures 2 and 3):
  1. When $B < \lambda_1/2$, the function is injective and is exactly (search) LWE!
  2. When $B > \lambda_1/2$, the function is *no longer injective*
     - **Pop Quiz 1.** When does the function first become surjective?
  3. When $B \geq \mu(\Lambda)$, the function is surjective.
     - Here $\mu(\Lambda)$ denotes the covering radius of the lattice, which is the radius of the smallest sphere centered *on the lattice* such that the ambient space is covered.
       - **Exercise 1.** Show that $\mu(\Lambda) \leq \sqrt{n} \cdot \lambda_n$.
     - Note that at this point, any point $\bar{a} \in \mathbb{R}^n$ in the ambient space can be represented by a lattice point $\bar{y} \in \Lambda$ and an error vector $\bar{e}$ of norm at most B as $\bar{a} := \bar{y} + \bar{e}$.
       - **Pop Quiz 2.** Why is the above true?
  4. When $B \gg \mu(\Lambda)$, the function is *very surjective* and has many collisions. This is exactly the regime that for SIS.

- In today's lecture
  - We will show a worst-case to average case reduction for SIS, which is due to Ajtai (STOC-Ajtai96 STUB) and with improvements by Micciancio and Regev (FOCS-MicReg04 STUB). That is we will show that if a certain lattice problem (looking ahead, SIVP) is hard in the worst case, then SIS is hard on average.

This result is quite remarkable since it implies cryptography based on worst-case assumptions! Prior to this result (standard) cryptography was known only from average-case assumptions like integer factorisation and discrete-logarithm problem.
    - Key to proving this result is the notion of smoothing and *smoothing parameter*, which we will define in this lecture. Intuitively, it is the bound B above which the output of Ajtai's function is distributed *statistically-close* to uniform over $P(\bar{B})$.
- Worst-case to Average-case Reduction
  - Recall the definition of SIVP from Lecture 05:
    - **Definition 1 (SIVP$_\alpha$: Approximate SIVP, Definition 5 from Lecture 05).**
      - *Input.* A lattice $\Lambda := \Lambda(\bar{B})$ and an approximation factor $\alpha(\cdot)$
      - *Solution.* m linearly-independent lattice vectors $\bar{y}_1, \cdots, \bar{y}_m \in \Lambda$ such that $\|\bar{y}_i\| \leq \alpha(n) \cdot \lambda_n(\Lambda)$
  - The theorem we prove today is stated below: informally, it claims that the SIS distribution is hard on average as long as there *exists* one hard instance of SIVP (see Figure 4).
  - **Theorem 1 (Theorem 2.1 in [2]).** There is a (randomised) poly-time reduction from SIVP$_{\tilde{O}(n)}$ to average-case SIS for parameters

$$q = \Omega(n^2), m \approx n \log(q) \text{ and } B = O(\sqrt{m}).$$

  - *Proof.*
    - *A simple (but wrong) reduction [1].*
      - We first give a simplified reduction that conveys all the essential ideas, but is *not* technically sound. Later we will see what the issue are and how these can be fixed. On a high level, given the worst-case lattice $\Lambda = \Lambda(\bar{B})$, we want to construct a SIS matrix $\bar{A}$ encoding a (dual) lattice $\Lambda_q^\perp(\bar{A})$ in such a way that
        1. The entries of $\bar{A}$ are random; and
        2. Yet $\Lambda_q^\perp(\bar{A})$ still carries a "signal" about the original lattice Λ.
          - We can then feed $\bar{A}$ to the SIS solver to obtain a collision or short $\{-1, 0, 1\}$ vector, which we then exploit to solve SIVP (see Figure 5). So the question is how to generate $\bar{A}$? To this end, we return to our discussion about how the bound B affects the output distribution of generalised Ajtai's function (recall Figures 2 and 3). It can be shown that once the norm bound B is large enough, e.g.,

$$\|\bar{e}\| \approx \log(n) \cdot \mu(\Lambda),$$

            the output of Ajtai's function is distributed statistically-close to uniform distribution over $P(\bar{B})$. Recall that:

- **Definition 2.** The statistical distance between two (continuous) distributions D_1 and D_2 over Ω is defined as

$$\mathsf{SD}(D_1, D_2) := 1/2 \int_{x \in \Omega} |D_1(x) - D_2(x)| dx.$$

- Note that since $P(\bar{B})$ has measurable volume, it is possible to define uniform distribution over it.

- The minimum bound B where the above requirement holds is known as the *smoothing parameter* η of the lattice basis. The preceding discussion hints to the following natural way to sample a vector (from the ambient space) that is random, but still depends on Λ:

1. Sample a random lattice point $\bar{y}_i \in \Lambda$ and random error vector $\bar{e}_i$ of norm at most η to get an ambient vector

$$\bar{a}_i := \bar{y}_i + \bar{e}_i.$$

By a similar argument to above, $\bar{a}_i \in \mathbb{R}^n$ is uniformly distributed $\mathbb{R}^n$. (Alternatively, one could sample a random $\bar{a}_i$ from the ambient space and then find the unique representative modulo $\bar{y}_i$ modulo Λ.)

- Let us now see what happens if you call the SIS solver on $\bar{A} := (\bar{a}_1, \cdots, \bar{a}_m)$ to obtain a short vector $\bar{e}^* \in \{-1, 0, 1\}^m$ in $\bar{A}$'s kernel (i.e., a lattice point in the dual lattice $\Lambda_q^{\perp}(\bar{A})$), i.e.,

$$\bar{A}\bar{e}^* = \sum_{i=1}^{m} e_i^* \bar{a}_i = 0^n,$$

where the arithmetic is over $\mathbb{R}^n$. This implies

$$\sum_{i=1}^{m} e_i^* (\bar{y}_i + \bar{e}_i) = 0 \Leftrightarrow \sum_{i=1}^{m} e_i^* \bar{y}_i = -\sum_{i=1}^{m} e_i^* \bar{e}_i.$$

Now observe that $\sum_{i=1}^{m} e_i^* \bar{y}_i$

1. Is a point in Λ since $\bar{y}_i$s are all in Λ and $\bar{e}^*$ is an integer vector (note that this implies that $\sum_{i=1}^{m} e_i^* \bar{e}_i$ is also a lattice point although $\bar{e}_i$s are *not* in the lattice)
2. Has the same norm as $\sum_{i=1}^{m} e_i^* \bar{e}_i$, which is at most

$$\sum_{i=1}^{m} e_i^* \cdot \max_{i \in [1,m]} \|\bar{e}_i\| = \sqrt{m} \max_{i \in [1,m]} \|\bar{e}_i\| = \sqrt{m} \approx n\lambda_n.$$

- Thus we have managed to obtain a shortish vector in $\Lambda(\bar{B})$! The *partial* reduction is described below:
  - **Reduction 1.**
    - *Input:* worst-case lattice $\Lambda(\bar{B})$ and oracle access to SIS solver

- *Output:* a shortish lattice vector $\bar{y} \in \Lambda(\bar{B})$
- *Reduction:*
  1. For each $i \in [1, m \approx n \log(q)]$: sample a random lattice point $\bar{y}_i \in \Lambda$ and random error vector $\bar{e}_i$ of norm at most η to get an ambient vector

$$\bar{a}_i := \bar{y}_i + \bar{e}_i.$$

  2. Call SIS oracle on $\bar{A}$ to obtain a vector $\bar{e}^* \in \{-1, 0, 1\}^m$.
  3. Output $\bar{y} := \sum_{i=1}^m e_i^* \bar{y}_i$

- **Exercise 2 (Reduce SIVP$_\alpha$ to SIS).** Figure out how to extract the rest of the vectors to solve SIVP.
  - *Hint.* You will need to invoke the above reduction multiple times, improving the basis $\bar{B}$ each time using the output of the SIS solver. In particular, be wary of the oracle returning 0^n.

Further Reading

-

2026s-CS758-Lecture12

Topics

- Worst-case to average-case reduction for SIS, continued
- Gaussian distribution and smoothing parameter

Sources

1. The Short Integer Solutions Problem and Cryptographic Applications, by Micciancio
2. Advanced Topics in Cryptography: Lattices (Fall 2015), Lectures [11](Lecture 11) and [12](Lecture 11), by Vaikuntanathan
3. [Lattices in Computer Science](https://cims.nyu.edu/~regev/papers/average.pdf, Lecture 12 by Regev

Plan

- Recall/Plan Today
  - In the last lecture:
    - We were trying to prove
      - **Theorem 1 (Theorem 4 in 3).** There is a (randomised) poly-time reduction from SIVP$_{O(n^3)}$ to average-case SIS for parameters

$$q = 2^{2n} \text{ and } m = 4n^2.$$

        - Recall that we had claimed to prove for (generalised) SIS regime

$$q = \Omega(n^2), m \approx n\log(q) \text{ and } B = O(\sqrt{m}),$$

          but we will be proving a weaker theorem.
    - In particular we were trying to reduce SIVP to SIS, by using the SIS oracle to extract a short vector in Λ. Our first attempt was the following simple (but wrong) reduction:
      - **Reduction 1 (Reduction 1 in Lecture 11).**
        - *Input:* worst-case lattice $\Lambda(\bar{B})$ and oracle access to SIS solver that outputs a short $\{-1, 0, 1\}$ vector in the kernel of the matrix $\bar{A}$ it is invoked with
        - *Output:* a shortish lattice vector $\bar{y}^* \in \Lambda(\bar{B})$
        - *Reduction:*
          1. For each $i \in [1, m \approx n\log(q)]$: sample a random lattice point $\bar{y}_i \in \Lambda$ and random error vector $\bar{e}_i$ of norm at most smoothing parameter η to get an ambient vector

$$\bar{a}_i := \bar{y}_i + \bar{e}_i.$$

2. Call SIS oracle on $\bar{A}$ to obtain a (short) vector $\bar{e}^* \in \{-1,0,1\}^m$ such that $\bar{A}\bar{e}^* = 0$ (over reals).
3. Output $\bar{y}^* := \sum_{i=1}^m e_i^* \bar{y}_i = -\sum_{i=1}^m e_i^* \bar{e}_i$
    - Recall that we argued that $\bar{y}^*$ is a short lattice vector since $\bar{y}_i$s are lattice vectors, $\bar{e} \in \{-1,0,1\}^m$ and $\bar{e}_i$ are short by the choice of norm B.
  - What are the issues with the above attempt?
    1. The matrix $\bar{A}$ we constructed was over $\mathbb{R}^{n\times m}$ instead of $\mathbb{Z}_q^{n\times m}$, and we claimed that $\bar{A}$ is "uniform" over $\mathbb{R}^{n\times m}$ (what does this even mean?).
    2. Turns out that we do not know how to *analyse* the smoothing lemma when using uniform distribution.
- In today's lecture:
  - We will resolve these issues by:
    1. Using n-dimensional Gaussian distribution instead of uniform distribution to sample vectors $\bar{a}_i$.
    2. Restricting to $P(\bar{B})$ by reducing modulo the lattice and then quantising $P(\bar{B})$ to associate $\mathbb{Z}_q^n$ with it.
  - The key tool we will use is Fourier analysis, which we discuss in the next lecture.
- Gaussian Distribution and the Smoothing Parameter
  - *Gaussian Distribution.* We first define the Gaussian distribution.
    - **Definition 1 ([3]).** For any "centre" $\bar{c} \in \mathbb{R}^n$ and "width" (or "radius") $s > 0$, the n-dimensional (continuous) Gaussian distribution over $\mathbb{R}^n$ *centered* at $\bar{c}$ and *scaled* by s is defined by the Gaussian function

$$\rho_{\bar{c},s}(\bar{a}) := e^{-\pi\|(\bar{a}-\bar{c})/s\|^2},$$

      and the density function

$$D_{\bar{c},s}(\bar{a}) := \rho_{\bar{c},s}(\bar{a})/s^n.$$

    - By default we consider $\bar{c} = 0^n$ and $s = 1$ (see **Figure 1**), in either of which case we drop it from the subscript.
    - **Exercise 1.**
      1. Prove that the Gaussian function $\rho_{\bar{c},s}$ sums over $\mathbb{R}^n$ to s^n, which explains the normalising factor $1/s^n$ when defining the density function.
      2. What is the mean and variance of $D_{\bar{c},s}$?
    - **Exercise 2.** When $\bar{c} = 0^n$, show that a random vector sampled according to D_s has length at most $s\sqrt{n}$ with probability $1 - 2^{\Omega(n)}$.
  - *Smoothing parameter.* Informally, if one picks an error vector from a Gaussian distribution with width s at least as large as the smoothing parameter, and reduces it modulo the lattice, then the resulting distribution is almost uniform.
    - **Definition 2 (Definition 2.3 in [2]).** The smoothing parameter of a lattice $\Lambda(\bar{B})$ with error ε, denoted η_ε, is the smallest width s such that $D_s \bmod \Lambda$ is ε-

close in statistical distance to the uniform distribution over $P(\bar{B})$.

- Looking ahead, we are going to require the following key lemma, which upper bounds the SD between the two distributions in consideration in terms of measure of a related Gaussian distribution -- looking ahead, on its Fourier transform -- on the *dual lattice*.
  - **Lemma 1 (Smoothing lemma, Lemma 5 in [3]).** For any lattice $\Lambda = \Lambda(\bar{B})$ and any width $s > 0$,

    $$\text{SD}(U_{P(\bar{B})}, R_{s,P(\bar{B})}) \leq \rho_{1/s}(\Lambda^* \setminus \{0^n\})/2,$$

    where
    - $U = U_{P(\bar{B})}$ denote the uniform distribution over $P(\bar{B})$,
    - $R = R_{s,P(\bar{B})}$ the induced random distribution $\rho_s \bmod P(\bar{B})$
    - $\rho_{1/s}(\Lambda^* \setminus \{0^n\})$ is the Gaussian distribution of width $1/s$ restricted to the non-zero entries of the dual lattice Λ^*, i.e.,

      $$\sum_{\bar{x} \in \Lambda^* \setminus \{0^n\}} \rho_{1/s}(\bar{x}).$$

  - Given Lemma 1, it can be shown that if the width s of the Gaussian is sufficiently large, then the SD between U and R is negligible. In particular,
  - **Theorem 1 (Theorem 11 in [2]).** For a lattice $\Lambda = \Lambda(\bar{B})$, if $s > \sqrt{n} \cdot \lambda_n(\Lambda)$ then

    $$\text{SD}(U, R) \leq 2^{-n}.$$

  - *Proof.* See [2]
- What do we know about the value of η_ε?
  - **Claim 1 (Corollary 8, Claim 9 and Corollary 10 in [3]).** For $\varepsilon < 1/100$,

    $$\lambda_n(\Lambda)/n \leq \eta_\varepsilon(\Lambda) \leq \sqrt{n}/\lambda_1(\Lambda^*).$$

    For $\varepsilon \geq 1/2^{n-1}$,

    $$\eta_\varepsilon(\Lambda) \leq \sqrt{n} \cdot \lambda_n(\Lambda).$$

- *The reduction [2].*
  - Recall from earlier discussion we fix Reduction 1 by:
    1. Using n-dimensional Gaussian distribution to sample vectors $\bar{e}_i$.
    2. Restricting to $P(\bar{B})$ by reducing $\bar{e}_i$ modulo the lattice and then quantising $P(\bar{B})$ to map it to $\mathbb{Z}_q^n$. To be precise:
       1. We first partition $P(\bar{B})$ into q^n equal "sub-parallelepipeds" (for $q = 2^{2n}$ as stated in the theorem). These sub-parallelepipeds can be

thought to correspond to the lattice $\Lambda(\frac{1}{q}\bar{B})$ obtained by scaling Λ down by a factor $1/q$.

2. Then we use each sub-parallelepiped to represent an element in $\mathbb{Z}_q^n$ (see **Figure 2**).

- **Pop-quiz 1.** Given a vector $\bar{a}' \in P(\bar{B})$, how can one find the *index* $\bar{a} \in \mathbb{Z}_q^n$ of the sub-parallelepiped it belongs to?
- *Solution.* We proceed similar to reducing an ambient vector modulo a lattice basis. Thus, there are two ways:
  1. Find the representative $\bar{a}$ of $\bar{a}'$ in $\Lambda(\frac{1}{q}\bar{B})$. That is, if

     $$\bar{a}' = \frac{1}{q}\bar{B}\bar{x}'$$

     for some $\bar{x}' \in \mathbb{R}^n$, then $\bar{a}$ can be obtained by simply rounding $\bar{a}'$ down, i.e., $a_i := \lfloor a_i' \rfloor$ for all $i \in [n]$.
  2. Multiply with the dual basis of $(\frac{1}{q}\Lambda)^*$, which is $q(\bar{B}^{-1})^t$, and then round:

     $$\bar{a} := \lfloor q \cdot \bar{B}^{-1}\bar{a}' \rceil$$

- The reduction is described below in detail.
  - **Reduction 2 ([2]).**
    - *Input:* worst-case lattice $\Lambda := \Lambda(\bar{B})$ and oracle access to SIS solver
    - *Output:* a shortish lattice vector $\bar{y} \in \Lambda(\bar{B})$
    - *Reduction:*
      1. For each $i \in [1, m]$
         1. Sample $\bar{e}_i$ from n-dimensional Gaussian D_s with $s \geq \eta_\varepsilon$.
         2. Reduce the error modulo the lattice, i.e.,

            $$\bar{a}_i' := \bar{e}_i \bmod \Lambda.$$

         3. Round $\bar{a}_i'$ to its sub-parallelepiped $\bar{a}_i$, i.e., Set

            $$\bar{a}_i := \lfloor q \cdot \bar{B}^{-1}\bar{a}_i' \rceil.$$

         4. Compute

            $$\bar{y}_i := \frac{1}{q}\bar{B}\bar{a}_i \in \frac{1}{q}\Lambda,$$

            which is the representative of $\bar{a}_i'$ in $\frac{1}{q}\Lambda$.
      2. Call SIS solver on $(\bar{a}_1, \cdots, \bar{a}_m)$ to obtain a short vector $\bar{e}^* := (e_1^*, \cdots, e_m^*) \in \{-1, 0, 1\}^m$ such that $\bar{A}\bar{e}^* = 0 \bmod q$.
         - If SIS solver fails, Reduction 2 *fails*.
      3. Output $\bar{y}^* := \sum_{i=1}^m e_i^*(\bar{e}_i - \bar{a}_i' + \bar{y}_i)$.

- **Theorem 2.** Reduction 2 outputs a shortish vector in Λ with overwhelming probability.
- *Proof sketch.* We split the proof into the following claims.
  - **Claim 2 (Claim 11 in [3]).** If Reduction 2 does not fail then it outputs a lattice vector.
    - **Exercise 3.** Prove Claim 1
  - **Claim 3 (Claim 12 in [3]).** If Reduction 2 does not fail then it outputs a shortish vector of length at most $O(n^3 \cdot \lambda_n(\Lambda))$ with overwhelming probability.
  - *Proof sketch.* Note that by triangle inequality and the fact that $\bar{e}_i$ has $\{-1, 0, 1\}$-entries

    $$\|\bar{y}^*\| \leq \sum_{i=1}^{m} \|e_i\| + \sum_{i=1}^{m} \|(\bar{y}_i - \bar{a}_i')\|.$$

    The term on the left can be bounded by Exercise 2 to be at most $4m\sqrt{n} \cdot \eta_\varepsilon$ with overwhelming probability. The term on the right can be bounded to be at most η_ε using the fact that the two vectors lie in the same sub-parallelepiped, and thus the difference of their length is bounded by the covering radius scaled down to sub-parallelepiped.
  - **Claim 4 (Claim 13 in [3]).** Reduction succeeds with probability at least $1/2n^{c_0}$ for a constant c_0.
  - *Proof sketch.* This is essentially a corollary to Theorem 1: since the SIS solver cannot tell the difference between honest SIS matrices and the matrices sampled as in Reduction 2, it is unlikely to fail.
  - **Claim 5 (Lemma 11 in [3]).** Reduction's output is a non-zero lattice point.
  - *Proof sketch.* Given $\bar{a}_i$s, the SIS solver can deduce $\bar{y}_i$s and $\bar{a}_i'$s to within good approximation. But there is still a lot of uncertainty about the value of $\bar{e}_i$s.

Further Reading

-

2026s-CS758-Lecture13

Topics

- Basic Fourier analysis
- Smoothing lemma

Sources

1. Lattices in Computer Science, Lecture 10, by Regev
2. Lattices in Computer Science, Lecture 12, by Regev

Plan

- Recall/Plan Today
  - In the last lecture:
    - We reduced approximate SIVP over a lattice $\Lambda = \Lambda(\bar{B})$ to SIS using Gaussian distribution $D_{\bar{c},s}$ and smoothing parameter η_ε.
    - In particular, we saw that if the distribution $R = R_{s,P(\bar{B})}$ induced by reducing D_s modulo the fundamental parallelepiped $P(\bar{B})$ is statistically-close to uniform distribution $U = U_{P(\bar{B})}$ over $P(\bar{B})$ then a random SIS instance $\bar{A} \in \mathbb{Z}_q^{n \times m}$ can be generated such that a short vector in its kernel can be used to extract a shortish vector in Λ. The corresponding lattice $\Lambda_q^\perp(\bar{A})$ is essentially a rounded, scaled down Λ, that is embedded in $P(\bar{B})$.
    - We then claimed that when $s \geq \eta_\varepsilon \geq \sqrt{n} \cdot \lambda_n(\Lambda)$, R is statistically close to U.
  - In today's lecture:
    - We will complete the analysis of the reduction by showing:
      - **Theorem 1 (Theorem 11 in [1]).** For any lattice $\Lambda = \Lambda(\bar{B}) \subset \mathbb{R}^n$, if $s \geq \sqrt{n} \cdot \lambda_n(\Lambda)$, then

$$\mathrm{SD}(U, R) \leq 2^{-n}.$$

    - As we will see from its proof, Theorem 1 boils down to the following "smoothing lemma", which relates the statistical distance above to the Gaussian mass over the dual lattice.
      - **Lemma 1 (Lemma 1 in Lecture 12).** For any lattice $\Lambda = \Lambda(\bar{B})$ and any width $s > 0$,

$$\mathrm{SD}(U_{P(\bar{B})}, R_{s,P(\bar{B})}) \leq \rho_{1/s}(\Lambda^* \setminus \{0^n\})/2,$$

        where $\rho_{1/s}(\Lambda^* \setminus \{0^n\})$ is the Gaussian distribution of width $1/s$ restricted

to the non-zero entries of the dual lattice Λ^*, i.e.,

$$\sum_{\bar{x} \in \Lambda^* \setminus \{0^n\}} \rho_{1/s}(\bar{x}).$$

- *Proof of Theorem 1.*
  - We will use the following lemma (without a proof).
    - **Lemma 2.** For any lattice $\Lambda \subset \mathbb{R}^n$, if $\lambda_1(\Lambda) > \sqrt{n}$, then $\rho_1(\Lambda \setminus \{0^n\}) \leq 2^{-n+1}$.
  - The lemma states that if the minimum distance of a lattice is at least $\sqrt{n}$, then the Gaussian distribution D_1 of width 1 restricted to the non-zero entries of the lattice has very small mass. Note that this is not surprising since the expected length of a vector sampled D_1 is $\sqrt{n}$, and it can be show that the longer lattice points only have a small mass.
  - We prove Theorem 1 using trade-offs between primal and dual lattices, in particular $\lambda_n(\Lambda)\lambda_1(\Lambda^*) \geq 1$. To this end, consider the scaled dual lattice $s\Lambda^*$. Note that

    $$\lambda_1(s\Lambda^*) = s\lambda_1(\Lambda^*) \geq \sqrt{n}\lambda_n(\Lambda)\lambda_1(\Lambda^*) \geq \sqrt{n},$$

    where the first equality follows by our assumption on s and the last inequality follows by Problem 1 in Assignment 2.
  - It follows by Lemma 2 that

    $$\rho_1(s\Lambda^* \setminus \{0^n\}) = \rho_{1/s}(\Lambda^* \setminus \{0^n\}) \leq 2^{-n+1},$$

    and the theorem follows by smoothing lemma.
  - In the rest of the lecture, we will prove smoothing lemma using Fourier analysis as a bridge between the primal and dual lattice.

- Fourier Analysis
  - *Fourier Transform.* Recall that a Fourier transform is a change of basis/representation, from "evaluation domain" to "function domain". For examples:
    - **Example 1.**
      1. Consider a audio signal. The standard way to think about audio signals is in terms of its amplitude over time, i.e., as a function $f(t)$ where e.g., $f(10)$ tells you the amplitude at $t = 10$. This is the so-called time domain representation. Alternatively, one can think of an audio signal also in terms of the constituent sine waves, i.e., as a function $\hat{f}(\nu)$, where $\hat{f}(10)$ tells you the amplitude of sine wave of 10Hz (see **Figure 1**). This is the so-called frequency domain representation. Time and frequency domain correspond to evaluation and function domains, respectively.
      2. Consider univariate polynomials (say, over $\mathbb{Z}_p$) of degree d. The standard way to think about it is via *coefficients* (of constituent monomials), i.e.,

$$p(x) = \sum_{i=0}^{d} c_i x^i.$$

This is representation in so-called monomial basis. Alternatively, one can think of a polynomial $p(x)$ also in terms of its *evaluations* $y_0, \cdots, y_d$ at (say) $0, \cdots, d \in \mathbb{Z}_p$. This is representation in so-called Lagrange basis. Representation in Lagrange and monomial bases correspond to evaluation and function domains, respectively.

- Fourier transform takes *any* complex function f in the evaluation domain and represents it in terms of constituent (basis) functions, viz., $e_y(x) := e^{-2\pi i y x}$. Let us define Fourier transform for general n-dimensional complex-valued functions.
  - **Definition 3 ([1]).** For any "well-behaved" function

    $$f : \mathbb{R}^n \to \mathbb{C},$$

    the n-dimensional Fourier transform

    $$\hat{f} : \mathbb{R}^n \to \mathbb{C}$$

    is defined as

    $$\hat{f}(\bar{y}) := \int_{\bar{x} \in \mathbb{R}^n} f(\bar{x}) e^{-2\pi i \langle \bar{x}, \bar{y} \rangle} d\bar{x}.$$

    The inverse Fourier transform is defined as

    $$f(\bar{x}) := \int_{\bar{y} \in \mathbb{R}^n} \hat{f}(\bar{y}) e^{2\pi i \langle \bar{x}, \bar{y} \rangle} d\bar{y}.$$

    - A function for us is well-behaved (e.g.) if $\int_{\mathbb{R}^n} |f(\bar{x})| dx < \infty$, i.e., has bounded "$\ell_1$ norm".
- **Example 2 (Examples 1 and 2 in [1]).**
  1. Consider the (modded) threshold function

     $$f_a(x) := \begin{cases} 1 & \text{if } |x| < a \\ 0 & \text{otherwise.} \end{cases}$$

     Then its Fourier transform is (see **Figure 2**)

     $$\hat{f}_a(y) = \int_{-a}^{a} e^{-2\pi i x y} dx = \frac{e^{-2\pi i a y} - e^{2\pi i a y}}{-2\pi i y} = \frac{\sin(2\pi a y)}{y\pi}.$$

     - What does the Fourier transform tell us? The dominant component is $e^{-2\pi i x}$ with amplitude $\hat{f}_a(1) = \sin(2\pi a)/\pi$, while the rest of the components are dampened by the y in the denominator.
  2. Consider the 1D centered Gaussian function of width $s > 0$, i.e.,

     $$\rho_s(x) = e^{-\pi (x/s)^2}.$$

- **Exercise 1.** Show that the Fourier transform of the 1D Gaussian function is itself, up to scaling, i.e,

$$\hat{f}(y) = se^{-\pi(ys)^2} = s\rho_{1/s}(y).$$

Extend to the n-dimensional case.

- In the exercise below, you will show that Fourier Transform satisfies interesting properties such as linearity.
- **Exercise 2 (Theorem 1 in [1]).**
  1. Show that Fourier transform is a linear operation, i.e., for any well-behaved functions f and g, $\widehat{f+g} = \hat{f} + \hat{g}$.
  2. Show that the Fourier transform of the translation of function f is a phase-change of $\hat{f}$: for any shift $\bar{s} \in \mathbb{R}^n$, if

$$f_{\bar{s}}(\bar{x}) := f(\bar{x} + \bar{s})$$

then

$$\hat{f}_{\bar{s}}(\bar{y}) := \hat{f}(\bar{y}) \cdot e^{2\pi i \langle \bar{y}, \bar{s} \rangle}$$

- *Fourier series: the 1D case.* Since we work with lattices, which are *periodic* objects, we need to talk about Fourier *series*, which is a special case of Fourier Transform. Let us first formally define periodic functions.
  - **Definition 4 (periodic functions).** A 1D complex function $f : \mathbb{R} \to \mathbb{C}$ is said to have period λ if

$$f(x + \lambda\mathbb{Z}) = f(x).$$

  - **Example 3.**
    1. $f(x) = \sin(2\pi x)$ is 1-periodic.
    2. $f(x) = \cos(\pi x)$ is 2-periodic
  - **Definition 5 ([1]).** The Fourier *series* of any "well-behaved" function

$$f : \mathbb{R} \to \mathbb{C},$$

with period λ is the function

$$\hat{f} : \frac{1}{\lambda}\mathbb{Z} \to \mathbb{C}$$

defined as

$$\hat{f}(y) := \frac{1}{\lambda} \int_{x=0}^{\lambda} f(x) e^{-2\pi i x y} dx.$$

$\hat{f}(y)$ is called the y-th Fourier coefficient. The inverse is defined as

$$f(x) := \sum_{y \in \frac{1}{\lambda}\mathbb{Z}} \hat{f}(y) e^{2\pi i x y}.$$

    - For (say) $\lambda = 1$, observe that $\hat{f}$ is defined only over $\mathbb{Z}$, and the intuitive reason for this is that when f is 1-periodic, only integer frequencies appear.
  - **Example 4.** Consider the periodic (modded) threshold function for $\lambda \in \mathbb{N}$ (see **Figure 3**)

$$f_{a,\lambda}(x) := \begin{cases} 1 & \text{if } |x| \bmod \lambda < a \\ 0 & \text{otherwise.} \end{cases}$$

    What is the Fourier series of $f_{a,\lambda}$?
  - A useful tool is the Poisson summation formula (PSF), which relates the *periodic sum* of a function f (e.g., over integers) to the periodic sum of its Fourier transform $\hat{f}$ (e.g., over integers).
  - **Theorem 1 (Theorem 7 in [1]).** For any "well-behaved" function f:

$$\sum_{x \in \lambda\mathbb{Z}} f(x) = \sum_{y \in \frac{1}{\lambda}\mathbb{Z}} \hat{f}(y).$$

  - *Proof.* See [1].
  - **Exercise 2.** Generalise Definitions 4 and 5 to n dimensions.
- *Fourier series over Lattices.*
  - Definitions 4 and 5 can be thought defining periodicity and Fourier series with respect to a *particular* lattice, viz., the integer lattice $\mathbb{Z}$. Thus the definition can be generalised to arbitrary lattice Λ as stated in Definitions 5 and 6 using dual lattice.
  - **Definition 5 (Λ-periodic functions).**
    - For a lattice $\Lambda \subset \mathbb{R}^n$, an nD complex-valued function $f : \mathbb{R}^n \to \mathbb{C}$ is said to be Λ-periodic if

$$f(\bar{x} + \Lambda) = f(\bar{x}).$$

    - **Example 5.** The "delta function" of a lattice

$$\delta_\Lambda(\bar{x}) := \begin{cases} 1 & \text{if } \bar{x} \in \Lambda \\ 0 & \text{otherwise} \end{cases}$$

      is Λ-periodic
  - **Definition 6.** Let $\Lambda = \Lambda(\bar{B})$ be a lattice and $f : \mathbb{R}^n \to \mathbb{C}$ be a Λ-periodic function. The *Fourier series* of f is the function

$$\hat{f} : \Lambda^* \to \mathbb{C}$$

    defined as

$$\hat{f}(\bar{y}) := \frac{1}{\det(\Lambda)} \int_{\bar{x} \in P(\bar{B})} f(\bar{x}) e^{-2\pi i \langle \bar{x}, \bar{y} \rangle} d\bar{x}.$$

The inverse is defined as

$$f(\bar{x}) := \sum_{y \in \Lambda^*} \hat{f}(\bar{y}) e^{2\pi i \langle \bar{x} \bar{y} \rangle}.$$

- **Exercise 3.** Let Λ be a lattice and $f : \mathbb{R}^n \to \mathbb{C}$ be a Λ-periodic function. Show that Fourier series of f is independent of its basis.

- For proving smoothing lemma, we will need PSF for lattices, stated below.
- **Theorem 1 (Lemma 13 in [1]).** Let Λ be a lattice and $f : \mathbb{R}^n \to \mathbb{C}$ be a well-behaved function. Then

$$\sum_{\bar{x} \in \Lambda} f(\bar{x}) = \frac{1}{\det(\Lambda)} \sum_{\bar{y} \in \Lambda^*} \hat{f}(\bar{y}).$$

- Proof of Smoothing Lemma
  - Recall that we need to calculate the SD between the following two density functions on $P(\bar{B})$:

$$U(\bar{x}) := 1/\det(\Lambda) = \det(\Lambda^*)$$

and

$$R(\bar{x}) := \sum_{\bar{a} : \bar{a} \bmod P(\bar{B}) = \bar{x}} \frac{1}{s^n} \rho_s(\bar{a}) = \frac{1}{s^n} \rho_s(\bar{x} + \Lambda),$$

where $\bar{x} + \Lambda$ is a coset of Λ. Note that $R(\bar{x})$ is Λ-periodic and thus its Fourier series $\hat{R} : \Lambda^* \to \mathbb{C}$ is well defined, it and can be shown to be

$$\hat{R}(\bar{y}) = \frac{1}{s^n} \det(\Lambda^*) \hat{\rho}_s(\bar{y}) = \det(\Lambda^*) \rho_{1/s}(\bar{y}).$$

  - **Exercise 4.** Show the above.
  - Using inversion formula $R(\bar{x})$ can be now represented as

$$\begin{aligned} R(\bar{x}) &= \sum_{\bar{y} \in \Lambda^*} \hat{R}(\bar{y}) e^{2\pi i \langle \bar{x}, \bar{y} \rangle} \\ &= \sum_{\bar{y} \in \Lambda^*} \det(\Lambda^*) \rho_{1/s}(\bar{y}) e^{2\pi i \langle \bar{x}, \bar{y} \rangle} \\ &= \det(\Lambda^*)(1 + \sum_{\bar{y} \in \Lambda^* \setminus \{0^n\}} \rho_{1/s}(\bar{y}) e^{2\pi i \langle \bar{x}, \bar{y} \rangle}), \end{aligned}$$

where the second equality follows by definition of $\hat{R}$. Thus

$$\begin{aligned}
\mathsf{SD}(U, R) &= \frac{1}{2} \int_{\bar{x} \in P(\bar{B})} |R(\bar{x}) - U(\bar{x})| d\bar{x} \\
&\leq \frac{1}{2} \mathrm{vol}(P(\bar{B})) \cdot \max_{\bar{x} \in P(\bar{B})} |R(\bar{x}) - \det(\Lambda^*))| \\
&= \frac{1}{2} \det(\Lambda) \cdot \det(\Lambda^*) \max_{\bar{x} \in P(\bar{B})} \left| \sum_{\bar{y} \in \Lambda^* \setminus \{0^n\}} \rho_{1/s}(\bar{y}) \cdot e^{2\pi i \langle \bar{y}, \bar{x} \rangle} \right| \\
&\leq \frac{1}{2} \sum_{\bar{y} \in \Lambda^* \setminus \{0^n\}} |\rho_{1/s}(\bar{y})| \\
&= \frac{1}{2} \rho_{1/s}(\Lambda^* \setminus \{0^n\}),
\end{aligned}$$

where the last inequality follows by:

- The fact that $\det(\Lambda) \cdot \det(\Lambda^*) = 1$;
- Triangle inequality; and
- Observing that $e^{2\pi i \langle \bar{y}, \bar{x} \rangle} > 1$.

Further Reading

- See this 3Blue1Brown video for a visual explanation of Fourier transform. See this 3Blue1Brown video for a visual explanation of Fourier series.